Air University

Department of Computer Science

Router-On-A-Stick Network Using GNS3

Members:

Haleema Abid

Samreen Shoukat

Rabya Hassan

Roohi Gul

Kanzul Eeman

Semester: BSCYS-3B

Submitted To: Sir Zohaib

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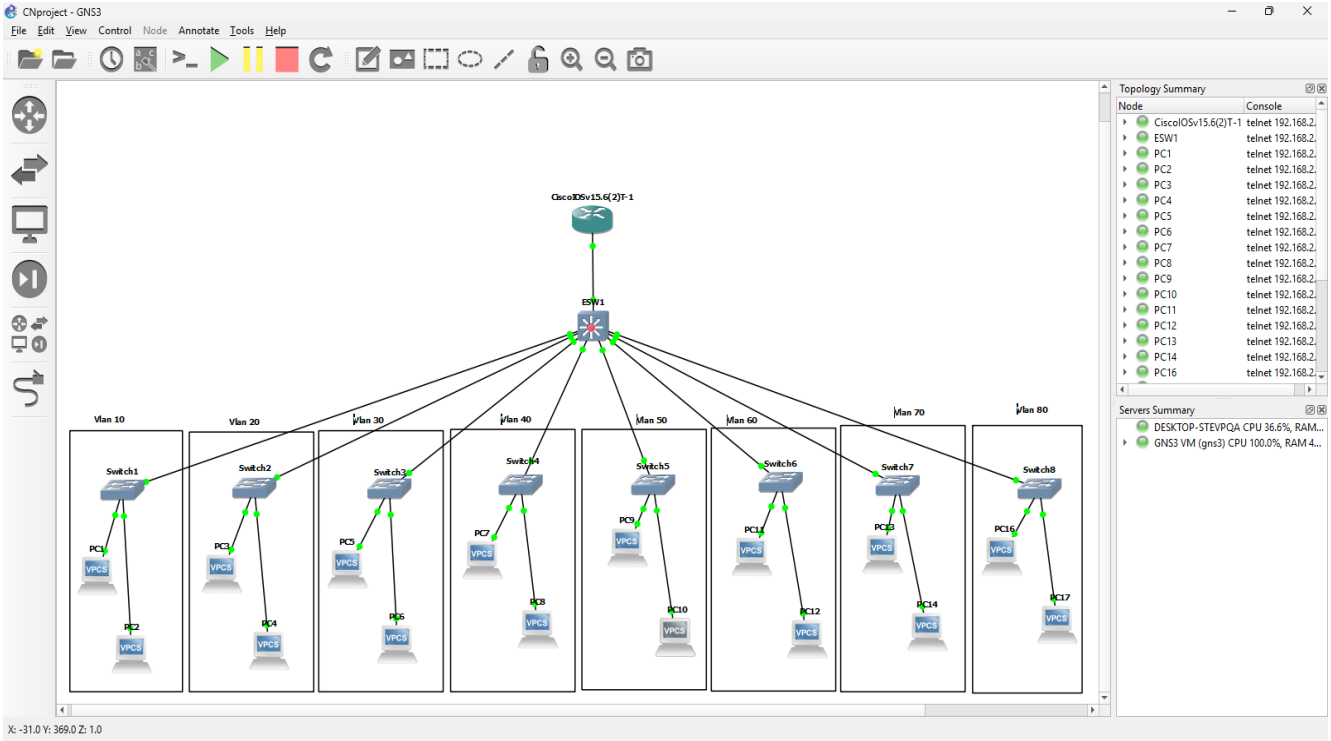
Network



Setup:

- **1 router**
  IOSv template: cisco-iosv
  Image: vios-adventerprisek9-m.vmdk.SPA.156-2
- **1 L3 switch**
  Image: c3640-a3js-mz.124-23
- **8 Ethernet switches(built in switch)**
- **16 vpcs**

VPCS Configuration:

VLAN 10:

- PC1: 192.168.10.2 gateway:192.168.10.1
- PC2: 192.168.10.3 gateway:192.168.10.1

```
PC1> ip 192.168.10.2/24 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.2 255.255.255.0 gateway 192.168.10.1

PC1>
```

```
PC2> ip 192.168.10.3/24 192.168.10.1
Checking for duplicate address...
PC2 : 192.168.10.3 255.255.255.0 gateway 192.168.10.1

PC2>
```

VLAN 20:

- PC1: 192.168.20.2 gateway:192.168.20.1
- PC2: 192.168.20.3 gateway:192.168.20.1

```
PC3> ip 192.168.20.2/24 192.168.20.1
Checking for duplicate address...
PC3 : 192.168.20.2 255.255.255.0 gateway 192.168.20.1

PC3>
```

```
PC4> ip 192.168.20.3/24 192.168.20.1
Checking for duplicate address...
PC4 : 192.168.20.3 255.255.255.0 gateway 192.168.20.1

PC4>
```

VLAN 30:

- PC1: 192.168.30.2 gateway:192.168.30.1
- PC2: 192.168.30.3 gateway:192.168.30.1

```
PC5> ip 192.168.30.2/24 192.168.30.1
Checking for duplicate address...
PC5 : 192.168.30.2 255.255.255.0 gateway 192.168.30.1

PC5>
```

```
PC6> ip 192.168.30.3/24 192.168.30.1
Checking for duplicate address...
PC6 : 192.168.30.3 255.255.255.0 gateway 192.168.30.1

PC6>
```

VLAN 40:

- PC1: 192.168.40.2 gateway:192.168.40.1
- PC2: 192.168.40.3 gateway:192.168.40.1

```
PC7> ip 192.168.40.2/24 192.168.40.1
Checking for duplicate address...
PC7 : 192.168.40.2 255.255.255.0 gateway 192.168.40.1

PC7>
```

```
PC8> ip 192.168.40.3/24 192.168.40.1
Checking for duplicate address...
PC8 : 192.168.40.3 255.255.255.0 gateway 192.168.40.1

PC8>
```

VLAN 50:

- PC1: 192.168.50.2 gateway:192.168.50.1
- PC2: 192.168.50.3 gateway:192.168.50.1

```
PC9> ip 192.168.50.2/24 192.168.50.1
Checking for duplicate address...
PC9 : 192.168.50.2 255.255.255.0 gateway 192.168.50.1

PC9>
```

```
PC10> ip 192.168.50.3/24 192.168.50.1
Checking for duplicate address...
PC10 : 192.168.50.3 255.255.255.0 gateway 192.168.50.1

PC10>
```

VLAN 60:

- PC1: 192.168.60.2 gateway:192.168.60.1
- PC2: 192.168.60.3 gateway:192.168.60.1

```
PC11> ip 192.168.60.2/24 192.168.60.1
Checking for duplicate address...
PC11 : 192.168.60.2 255.255.255.0 gateway 192.168.60.1

PC11>
```

```
PC12> ip 192.168.60.3/24 192.168.60.1
Checking for duplicate address...
PC12 : 192.168.60.3 255.255.255.0 gateway 192.168.60.1

PC12>
```

VLAN 70:

- PC1: 192.168.70.2 gateway:192.168.70.1
- PC2: 192.168.70.3 gateway:192.168.70.1

```
PC13> ip 192.168.70.2/24 192.168.70.1
Checking for duplicate address...
PC13 : 192.168.70.2 255.255.255.0 gateway 192.168.70.1

PC13>
```

```
PC15> ip 192.168.70.3/24 192.168.70.1
Checking for duplicate address...
PC15 : 192.168.70.3 255.255.255.0 gateway 192.168.70.1

PC15>
```

VLAN 80:

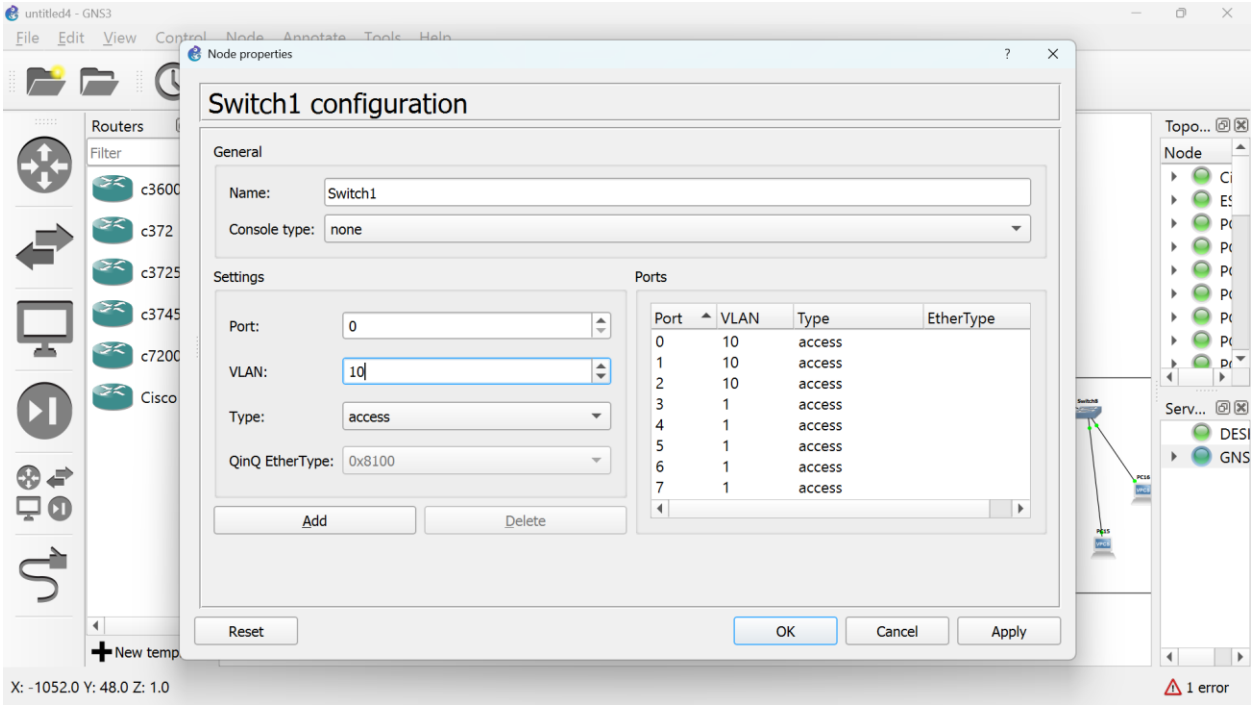
- PC1: 192.168.80.2 gateway:192.168.80.1
- PC2: 192.168.80.3 gateway:192.168.80.1

```
PC16> ip 192.168.80.2/24 192.168.80.1
Checking for duplicate address...
PC16 : 192.168.80.2 255.255.255.0 gateway 192.168.80.1

PC16>
```

```
PC17> ip 192.168.80.3/24 192.168.80.1
Checking for duplicate address...
PC17 : 192.168.80.3 255.255.255.0 gateway 192.168.80.1

PC17>
```

Note: Make sure to enter the command “save” after entering ip of each pc’s.

Ethernet Switch Configuration (built in):



Each of the 8 built-in switches requires the following port configuration:

- **Ports 0,1 & 2** (connected to PCs): **Access mode** - assigned to the switch's respective VLAN

Repeat this configuration for all 8 switches (Switch1 through Switch8), ensuring Port 0 is set to trunk and Ports 1-2 are set to access mode with the correct VLAN assignment

Ethernet Layer 3 Switch Setup

- **ESW1 VLAN Database Configuration**

```
ESW1(config)#exit
ESW1#
*Mar  1 00:12:30.099: %SYS-5-CONFIG_I: Configured from console by console
ESW1#vlan database
ESW1(vlan)#vlan 10 name vlan10
VLAN 10 added:
    Name: vlan10
ESW1(vlan)#vlan 20 name vlan20
VLAN 20 added:
    Name: vlan20
ESW1(vlan)#vlan 30 name vlan30
VLAN 30 added:
    Name: vlan30
ESW1(vlan)#vlan 40 name vlan40
VLAN 40 added:
    Name: vlan40
ESW1(vlan)#vlan 50 name vlan50
VLAN 50 added:
    Name: vlan50
ESW1(vlan)#vlan 60 name vlan60
VLAN 60 added:
    Name: vlan60
ESW1(vlan)#vlan 70 name vlan70
VLAN 70 added:
    Name: vlan70
ESW1(vlan)#vlan 80 name vlan80
VLAN 80 added:
    Name: vlan80
ESW1(vlan)#
```

The main switch (ESW1) has successfully created and configured a VLAN database containing 8 VLANs. Each VLAN is named according to its VLAN ID and has been added to the switch's VLAN database. The configuration includes:

- VLAN 10 (vlan10) - for Switch1
- VLAN 20 (vlan20) - for Switch2
- VLAN 30 (vlan30) - for Switch3
- VLAN 40 (vlan40) - for Switch4
- VLAN 50 (vlan50) - for Switch5
- VLAN 60 (vlan60) - for Switch6
- VLAN 70 (vlan70) - for Switch7
- VLAN 80 (vlan80) - for Switch8

All VLANs have been successfully added to the VLAN database on ESW1. This configuration enables proper VLAN segmentation across the network, with each of the 8 connected switches assigned to its own dedicated VLAN. The VLANs are then trunked to the router via port Fa0/8 for inter-VLAN routing.

- **ESW1 Access Port Configuration for VLAN Assignment**

```
ESW1(config)#interface fastEthernet 0/0
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 10
ESW1(config-if)#no shutdown
ESW1(config-if)#
*Mar  1 00:18:26.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ESW1(config-if)#
*Mar  1 00:18:28.419: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 20
ESW1(config-if)#no shutdown
ESW1(config-if)#
*Mar  1 00:19:31.183: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
ESW1(config-if)#
*Mar  1 00:19:33.099: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/2
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 30
ESW1(config-if)#no sh
*Mar  1 00:20:04.899: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/3
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 40
ESW1(config-if)#
*Mar  1 00:20:58.503: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/4
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 50
ESW1(config-if)#no shut
*Mar  1 00:22:03.043: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/5
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 60
ESW1(config-if)#
*Mar  1 00:22:38.627: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/6
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 70
ESW1(config-if)#
*Mar  1 00:23:24.511: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan70, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#interface fastEthernet 0/7
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 80
ESW1(config-if)#
*Mar  1 00:24:03.875: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan80, changed state to up
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#
```

The main switch (ESW1) has been configured with 8 FastEthernet access ports (Fa0/0 through Fa0/7), each assigned to a specific VLAN. Each port is set to access mode and has been brought up (no shutdown). The configuration includes:

- **Port Fa0/0**: Access mode, assigned to VLAN 10
- **Port Fa0/1**: Access mode, assigned to VLAN 20
- **Port Fa0/2**: Access mode, assigned to VLAN 30
- **Port Fa0/3**: Access mode, assigned to VLAN 40
- **Port Fa0/4**: Access mode, assigned to VLAN 50
- **Port Fa0/5**: Access mode, assigned to VLAN 60
- **Port Fa0/6**: Access mode, assigned to VLAN 70
- **Port Fa0/7**: Access mode, assigned to VLAN 80

All ports have been enabled and are operational, as confirmed by the LINEPROTO-5-UPDOWN system messages indicating each interface has come up.

- ## ESW1 Trunk Port Configuration

```
ESW1(config)#interface fastEthernet 0/8
ESW1(config-if)#switchport trunk encapsulation dot1q
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#
*Mar  1 00:28:06.223: %DTP-5-TRUNKPORTON: Port Fa0/8 has become dot1q trunk
ESW1(config-if)#no shutdown
ESW1(config-if)#exit
ESW1(config)#end
ESW1#wr
*Mar  1 00:28:37.699: %SYS-5-CONFIG_I: Configured from console by console
ESW1#write memory
Building configuration...
[OK]
ESW1#
```

The main switch (ESW1) has been configured with a trunk port (FastEthernet 0/8) to connect to the router. This port is configured to carry VLAN traffic using 802.1Q (dot1q) encapsulation, allowing multiple VLANs to pass through a single physical link. The configuration includes:

- **Port Fa0/8**: Trunk mode enabled with dot1q encapsulation
- **Encapsulation**: 802.1Q (dot1q) - standard VLAN tagging protocol
- **Status**: Interface is up and operational (LINEPROTO-5-TRUNKPORTON message confirms successful trunk activation)
- **No Shutdown**: Interface has been enabled

The trunk port successfully transitions to an active trunk state, as confirmed by the system message "Port Fa0/8 has become dot1q trunk." This trunk link carries all 8 VLANs (10, 20, 30, 40, 50, 60, 70, 80) from ESW1 to the router's Gi0/0 interface, enabling the router to perform inter-VLAN routing. The configuration has been saved to memory, ensuring persistence after device restart.

Router Subinterface Configuration for VLAN Routing

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.50
Router(config-subif)#encapsulation dot1Q 50
Router(config-subif)#ip address 192.168.50.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.60
Router(config-subif)#encapsulation dot1Q 60
Router(config-subif)#ip address 192.168.60.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.70
Router(config-subif)#encapsulation dot1Q 70
Router(config-subif)#ip address 192.168.70.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 192.168.80.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip routing
Router(config)#end
Router#writ
*Dec 13 14:26:15.293: %SYS-5-CONFIG_I: Configured from console by consolee
Router#write memory
Building configuration...
[OK]
```

The router has been configured with 8 subinterfaces on its Gi0/0 port, one for each VLAN. Each subinterface uses 802.1Q (dot1q) encapsulation and has been assigned a gateway IP address for its respective VLAN:

- **Gi0/0.10**: VLAN 10 gateway - 192.168.10.1/24
- **Gi0/0.20**: VLAN 20 gateway - 192.168.20.1/24
- **Gi0/0.30**: VLAN 30 gateway - 192.168.30.1/24
- **Gi0/0.40**: VLAN 40 gateway - 192.168.40.1/24
- **Gi0/0.50**: VLAN 50 gateway - 192.168.50.1/24
- **Gi0/0.60**: VLAN 60 gateway - 192.168.60.1/24
- **Gi0/0.70**: VLAN 70 gateway - 192.168.70.1/24
- **Gi0/0.80**: VLAN 80 gateway - 192.168.80.1/24

All subinterfaces are enabled (no shutdown) and IP routing is activated. This router-on-a-stick configuration enables inter-VLAN routing, allowing devices in different VLANs to communicate through the router via the single trunk link to ESW1.

Configuration Check

- **Switch check:**

```
ESW1#show interfaces status

Port      Name               Status       Vlan       Duplex Speed Type
Fa0/0     *** Unused for Lay connected    10         a-full   a-100 10/100BaseTX
Fa0/1     *** Unused for Lay connected    20         a-full   a-100 10/100BaseTX
Fa0/2                        connected    30         a-full   a-100 10/100BaseTX
Fa0/3                        connected    40         a-full   a-100 10/100BaseTX
Fa0/4                        connected    50         a-full   a-100 10/100BaseTX
Fa0/5                        connected    60         a-full   a-100 10/100BaseTX
Fa0/6                        connected    70         a-full   a-100 10/100BaseTX
Fa0/7                        connected    80         a-full   a-100 10/100BaseTX
Fa0/8                        connected    trunk      a-full   a-100 10/100BaseTX
Fa0/9                        notconnect   1            auto    auto 10/100BaseTX
Fa0/10                       notconnect   1            auto    auto 10/100BaseTX
Fa0/11                       notconnect   1            auto    auto 10/100BaseTX
Fa0/12                       notconnect   1            auto    auto 10/100BaseTX
Fa0/13                       notconnect   1            auto    auto 10/100BaseTX
Fa0/14                       notconnect   1            auto    auto 10/100BaseTX
Fa0/15                       notconnect   1            auto    auto 10/100BaseTX
Fa1/0                        notconnect   1            auto    auto 10/100BaseTX
Fa1/1                        notconnect   1            auto    auto 10/100BaseTX
Fa1/2                        notconnect   1            auto    auto 10/100BaseTX
Fa1/3                        notconnect   1            auto    auto 10/100BaseTX
Fa1/4                        notconnect   1            auto    auto 10/100BaseTX
```

- **Router check:**

```
Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.10      192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20      192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30      192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40      192.168.40.1    YES manual up                    up
GigabitEthernet0/0.50      192.168.50.1    YES manual up                    up
GigabitEthernet0/0.60      192.168.60.1    YES manual up                    up
GigabitEthernet0/0.70      192.168.70.1    YES manual up                    up
GigabitEthernet0/0.80      192.168.80.1    YES manual up                    up
GigabitEthernet0/1         unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2         unassigned      YES NVRAM  administratively down down
 --More--
```

Ping Testing

```
PC1> ping 192.168.80.3

84 bytes from 192.168.80.3 icmp_seq=1 ttl=63 time=98.815 ms
84 bytes from 192.168.80.3 icmp_seq=2 ttl=63 time=13.654 ms
84 bytes from 192.168.80.3 icmp_seq=3 ttl=63 time=6.213 ms
84 bytes from 192.168.80.3 icmp_seq=4 ttl=63 time=7.208 ms
84 bytes from 192.168.80.3 icmp_seq=5 ttl=63 time=3.576 ms

PC1>
```

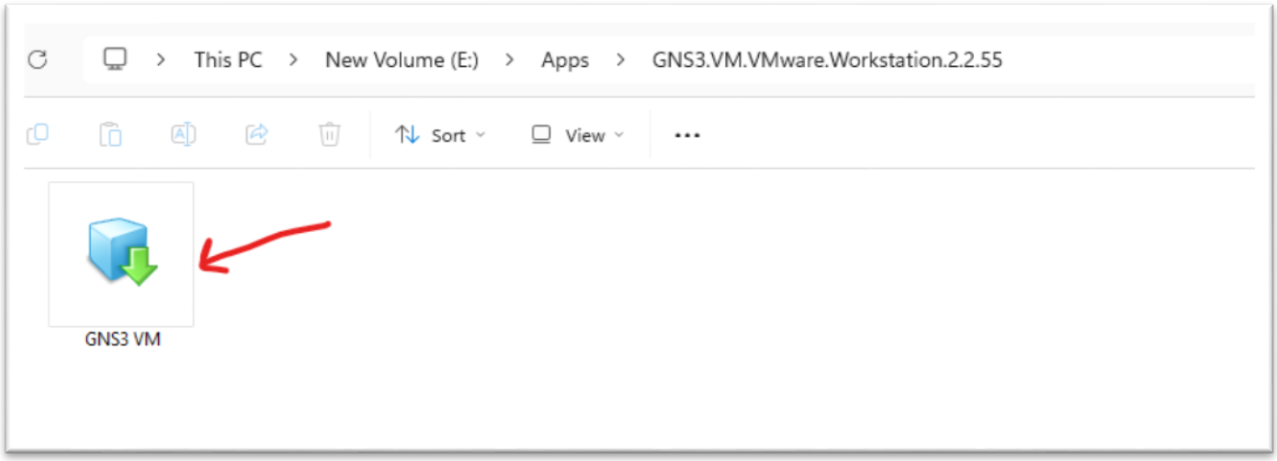
```
PC13> ping 192.168.50.2

84 bytes from 192.168.50.2 icmp_seq=1 ttl=63 time=79.540 ms
84 bytes from 192.168.50.2 icmp_seq=2 ttl=63 time=4.995 ms
84 bytes from 192.168.50.2 icmp_seq=3 ttl=63 time=4.730 ms
84 bytes from 192.168.50.2 icmp_seq=4 ttl=63 time=5.642 ms
84 bytes from 192.168.50.2 icmp_seq=5 ttl=63 time=7.006 ms

PC13>
```

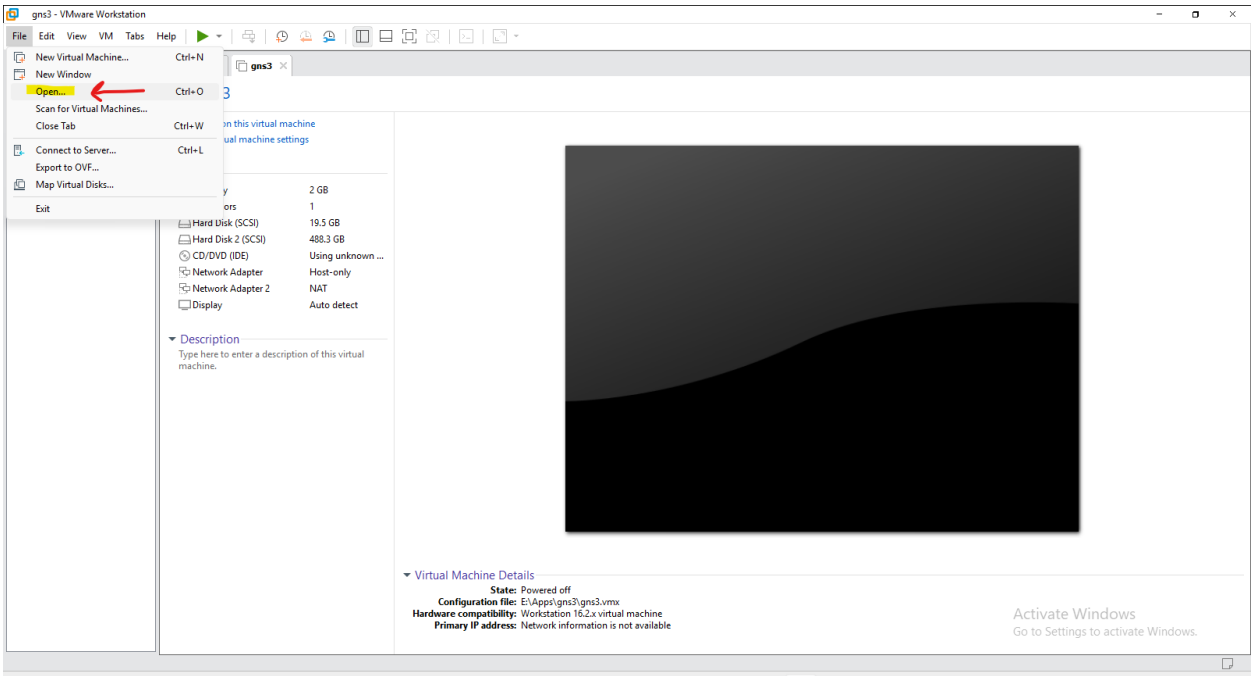
Pre-Requisites

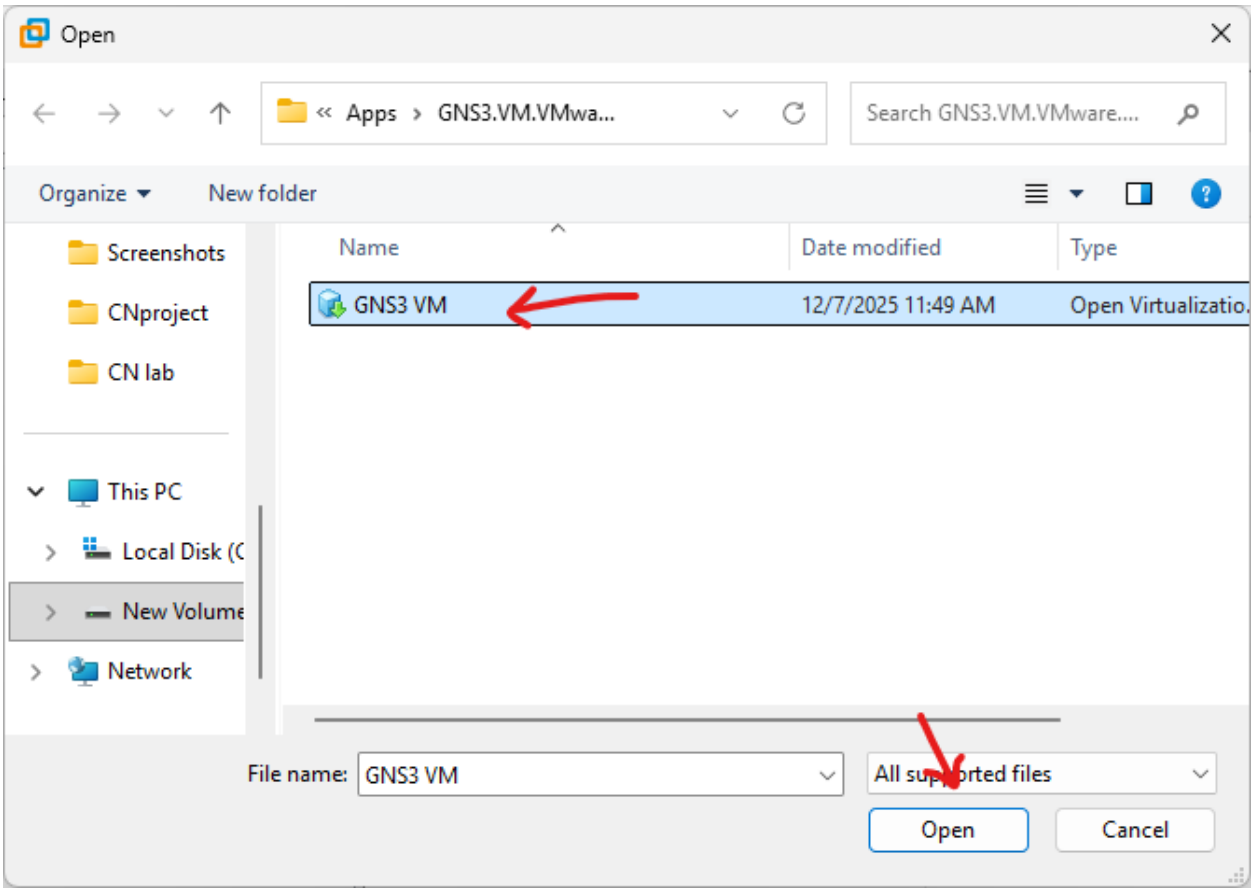
- ## Gns3 VM installation:

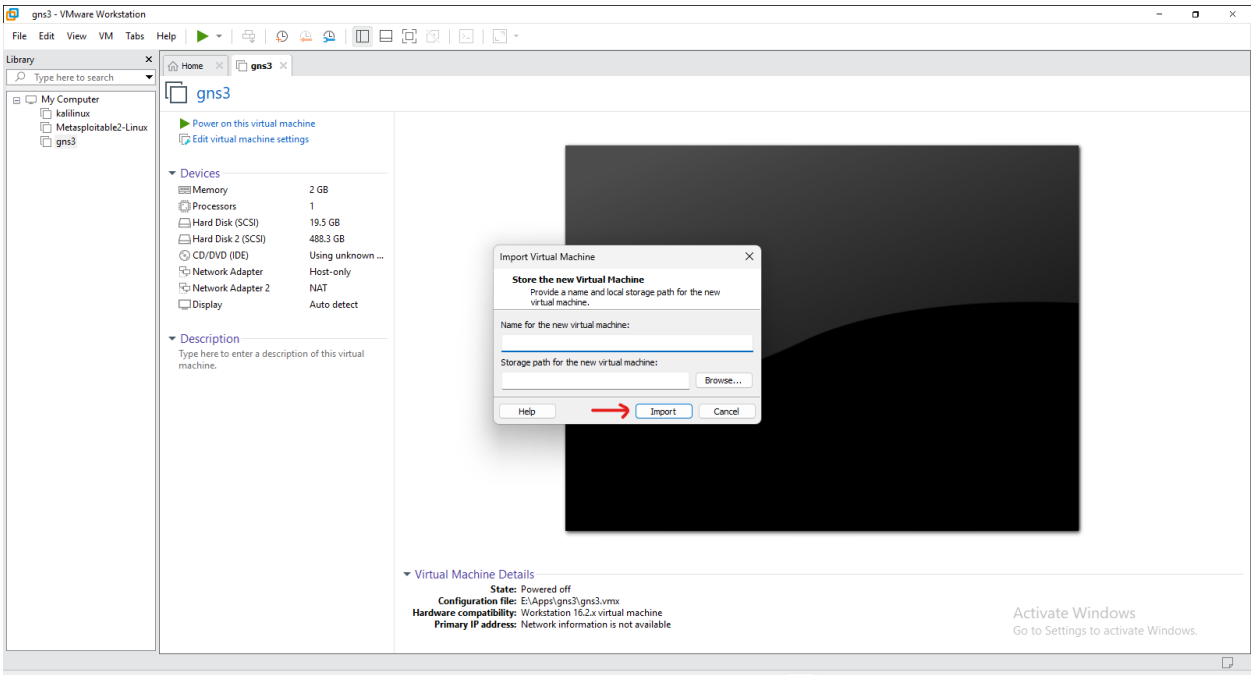
https://sourceforge.net/projects/gns-3/files/Releases/v2.2.15/GNS3.VM.VMware.Workstation.2.2.15.zip/download

Download this GNS3 virtual machine.





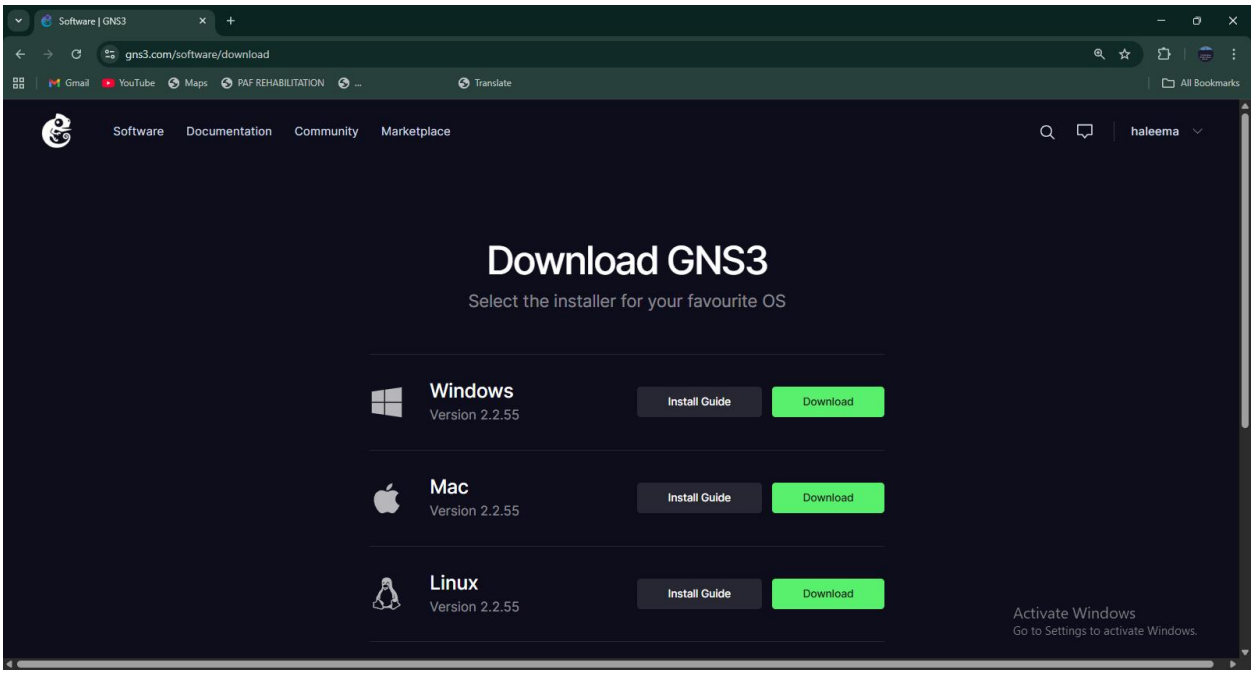




After naming the VM, import it and power on it.

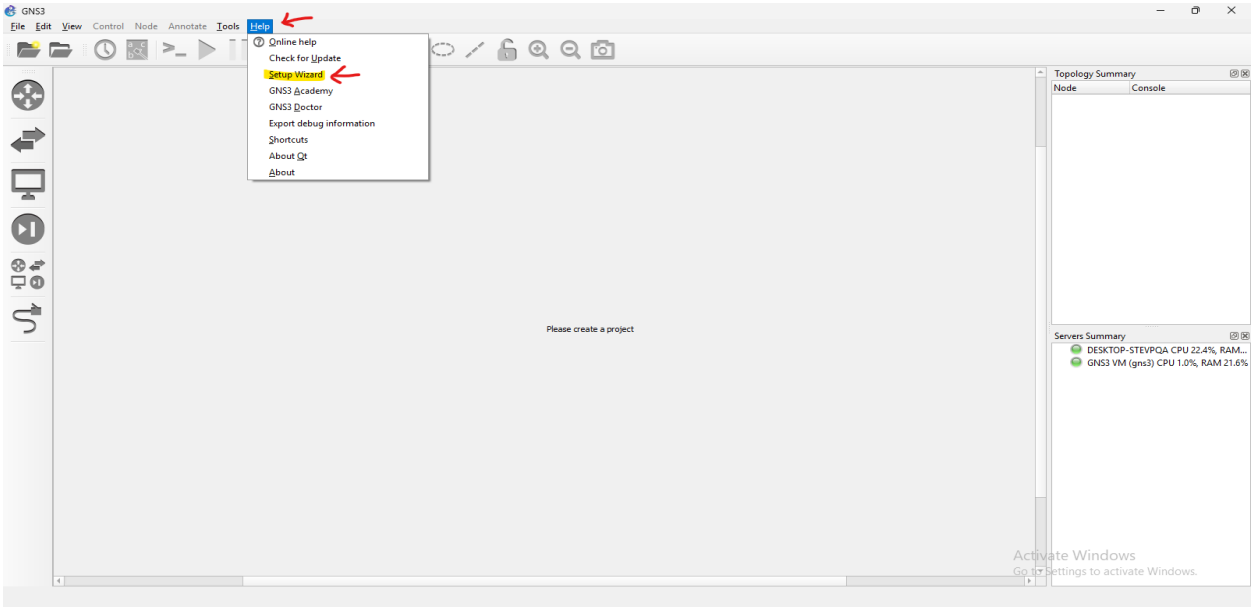
- ## Gns3 Sofware installation:

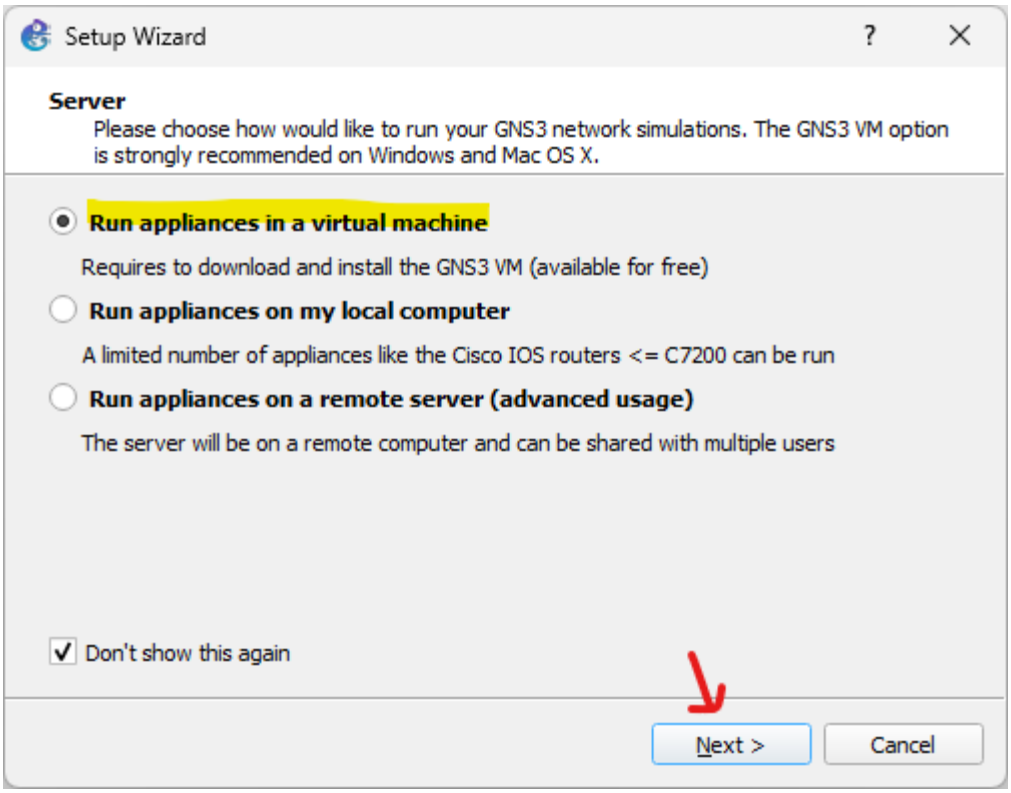
https://www.gns3.com/software/download

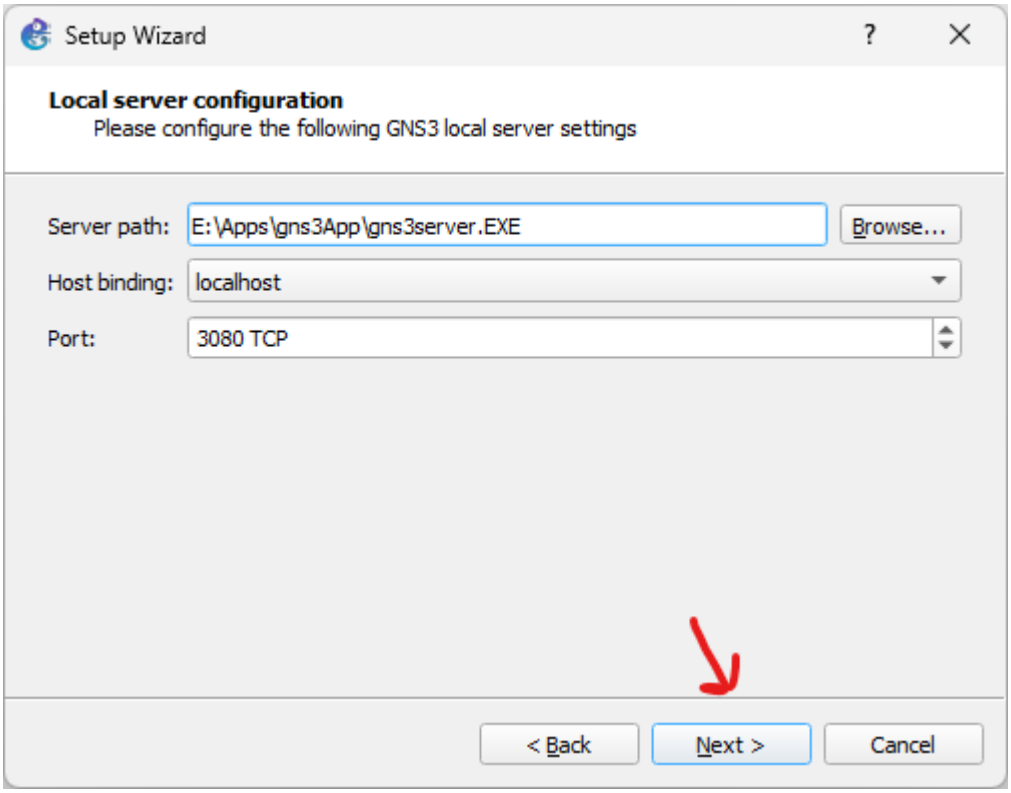


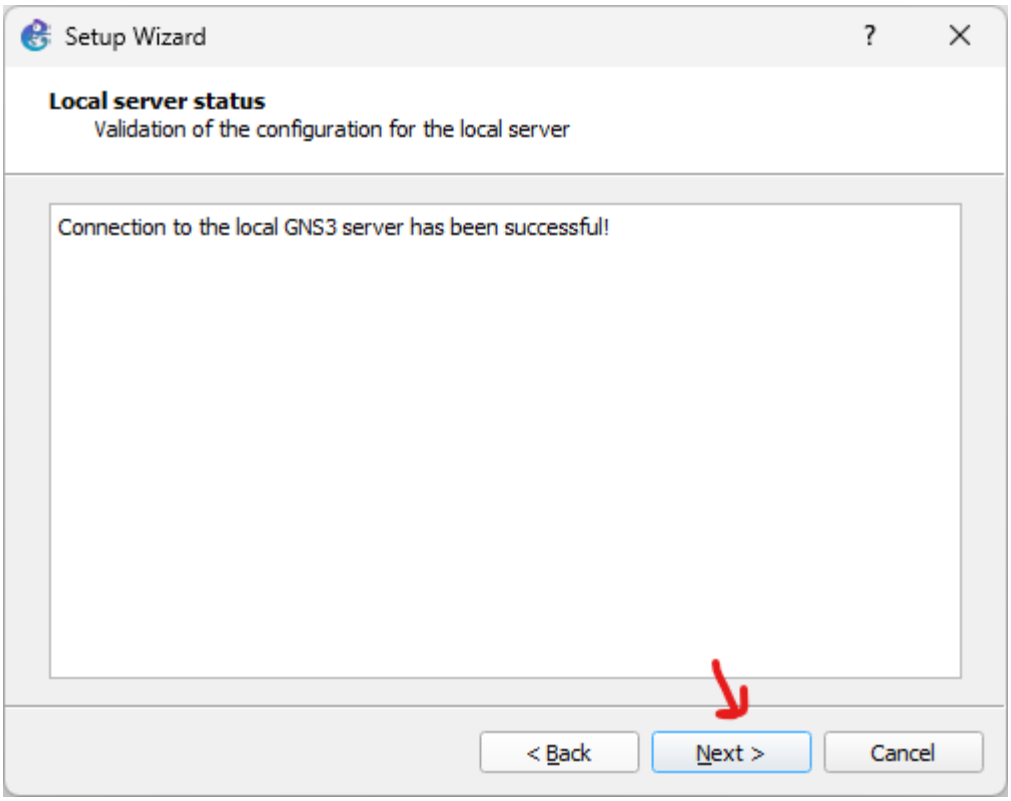
Download and install, gns3 software.

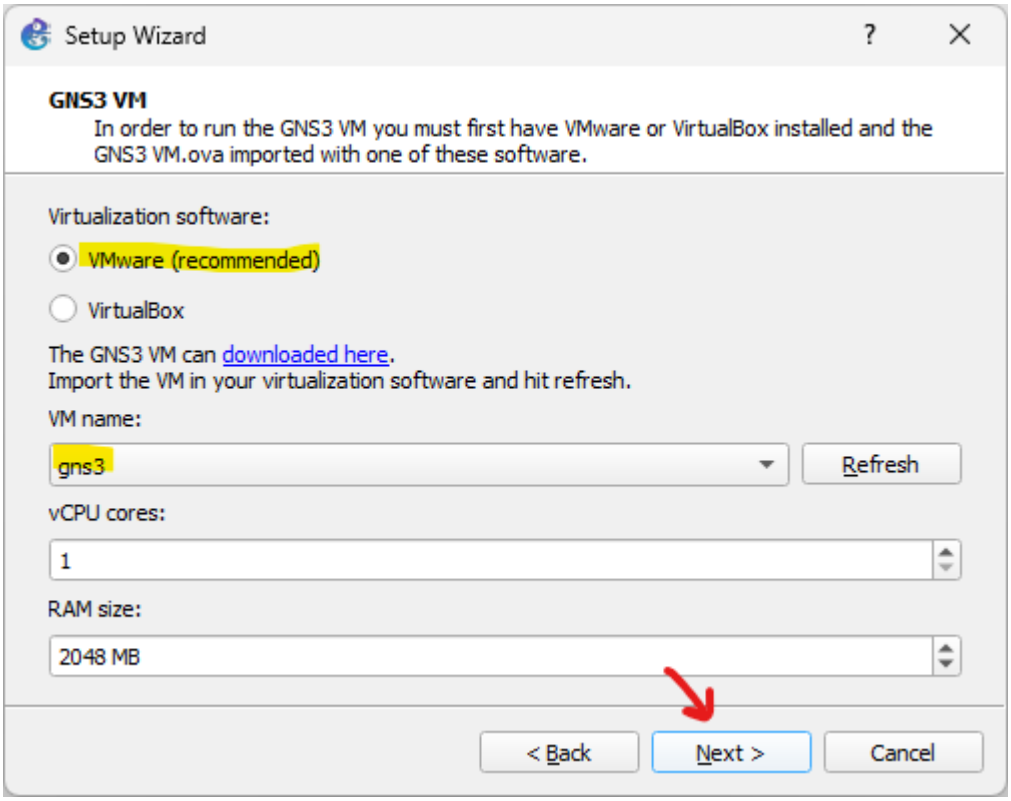
- ## Connecting GNS3 and VM:

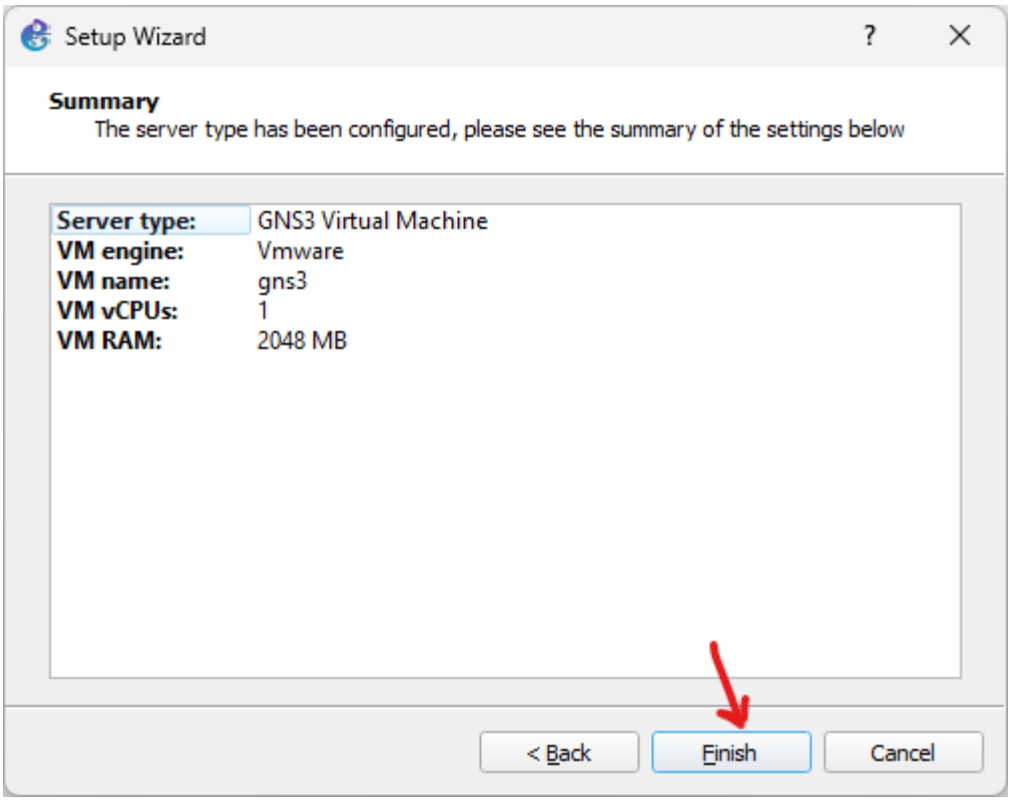



Setup Wizard
Server
Please choose how would like to run your GNS3 network simulations. The GNS3 VM option is strongly recommended on Windows and Mac OS X.
Run appliances in a virtual machine
Requires to download and install the GNS3 VM (available for free)
Run appliances on my local computer
A limited number of appliances like the Cisco IOS routers <= C7200 can be run
Run appliances on a remote server (advanced usage)
The server will be on a remote computer and can be shared with multiple users
Don't show this again
Next >
Cancel

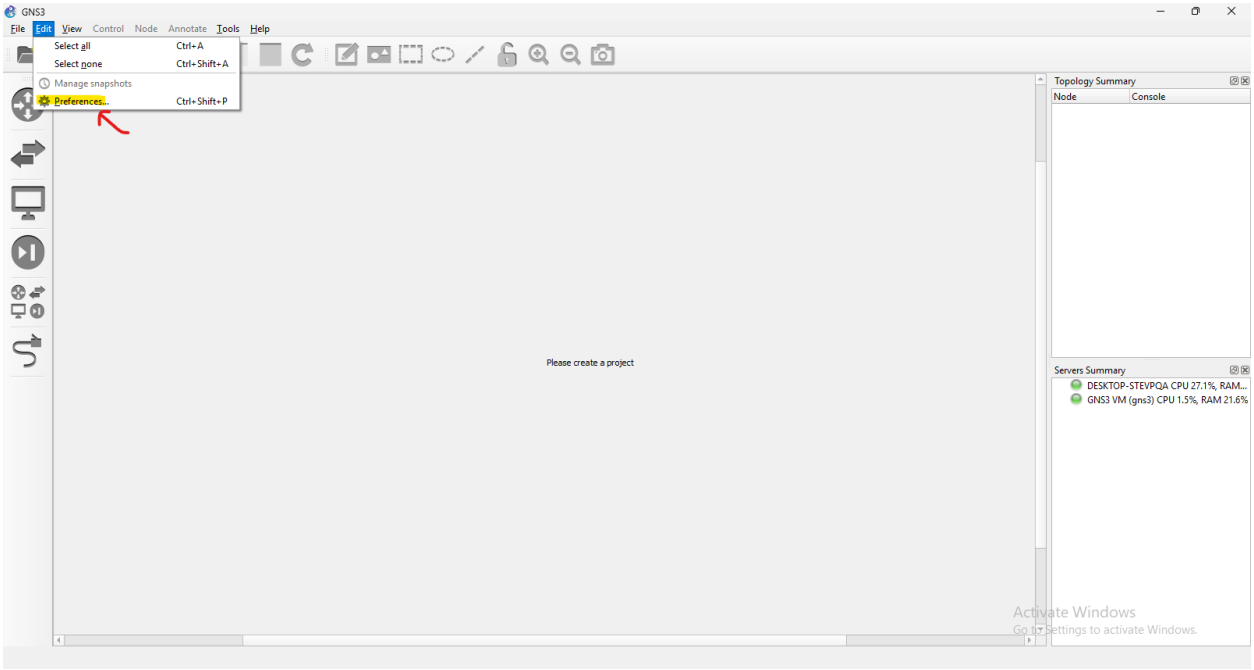

Setup Wizard
Local server configuration
Please configure the following GNS3 local server settings
Server path:
E:\Apps\gns3App\gns3server.EXE
Browse...
Host binding:
localhost
Port:
3080 TCP
< Back
Next >
Cancel

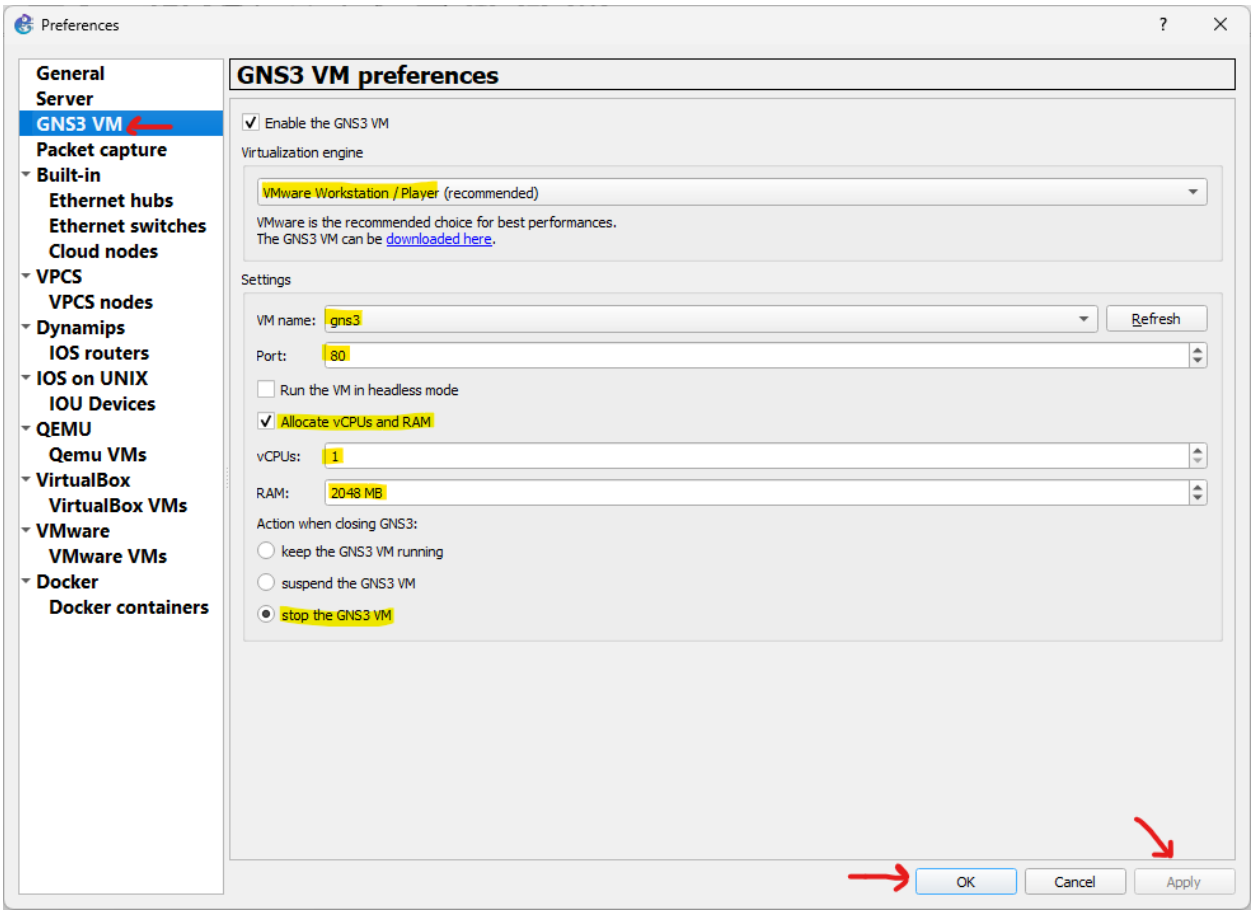

Setup Wizard
Local server status
Validation of the configuration for the local server
Connection to the local GNS3 server has been successful!
< Back
Next >
Cancel


Setup Wizard
GNS3 VM
In order to run the GNS3 VM you must first have VMware or VirtualBox installed and the GNS3 VM.ova imported with one of these software.
Virtualization software:
VMware (recommended)
VirtualBox
The GNS3 VM can downloaded here.
Import the VM in your virtualization software and hit refresh.
VM name:
gns3
Refresh
vCPU cores:
1
RAM size:
2048 MB
< Back
Next >
Cancel



Configuration GNS3:



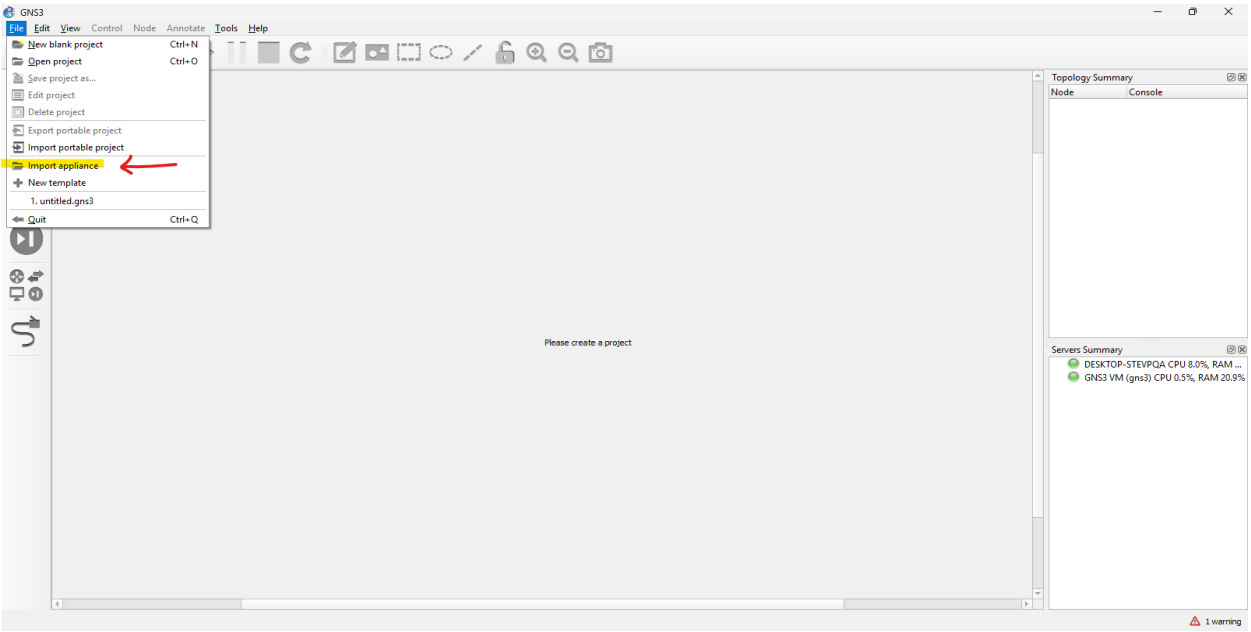

Preferences
General
Server
GNS3 VM
Packet capture
Built-in
Ethernet hubs
Ethernet switches
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containers
GNS3 VM preferences
Enable the GNS3 VM
Virtualization engine
VMware Workstation / Player (recommended)
VMware is the recommended choice for best performances.
The GNS3 VM can be downloaded here.
Settings
VM name:
gns3
Refresh
Port:
80
Run the VM in headless mode
Allocate vCPUs and RAM
vCPUs:
1
RAM:
2048 MB
Action when closing GNS3:
keep the GNS3 VM running
suspend the GNS3 VM
stop the GNS3 VM
OK
Cancel
Apply

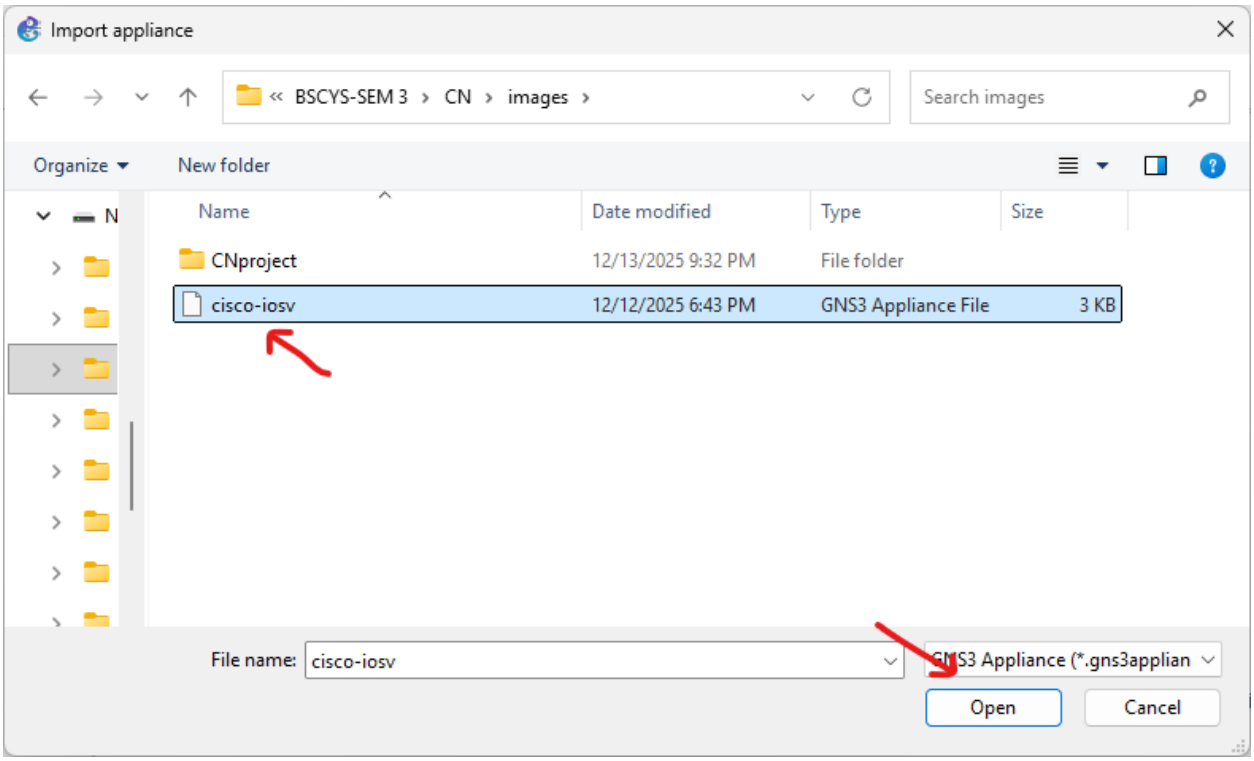
- ## Router And Switch Images:

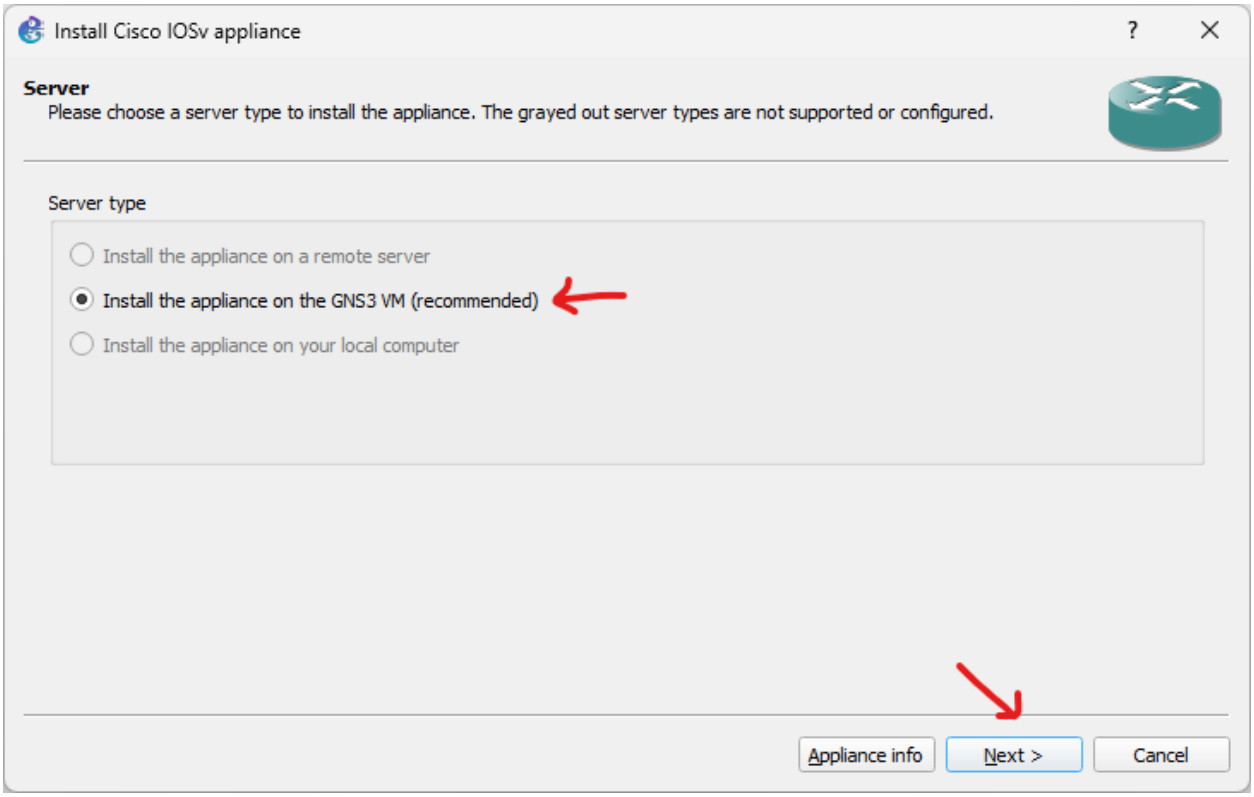
c3640-a3js-mz.124-23.image	12/13/2025 3:08 PM	IMAGE File	64,160 KB
cisco-iosv	12/12/2025 6:43 PM	GNS3 Appliance File	3 KB
cisco-iosv.gns3a.md5sum	12/12/2025 6:43 PM	MD5SUM File	1 KB
IOSv_startup_config	12/12/2025 6:42 PM	Disc Image File	1,024 KB
IOSv_startup_config.img.md5sum	12/12/2025 6:42 PM	MD5SUM File	1 KB
vios-adventerprisek9-m.vmdk.SPA.156-2	12/12/2025 6:41 PM	Perl Source File	125,440 KB
vios-adventerprisek9-m.vmdk.SPA.156-2....	12/12/2025 6:25 PM	MD5SUM File	1 KB

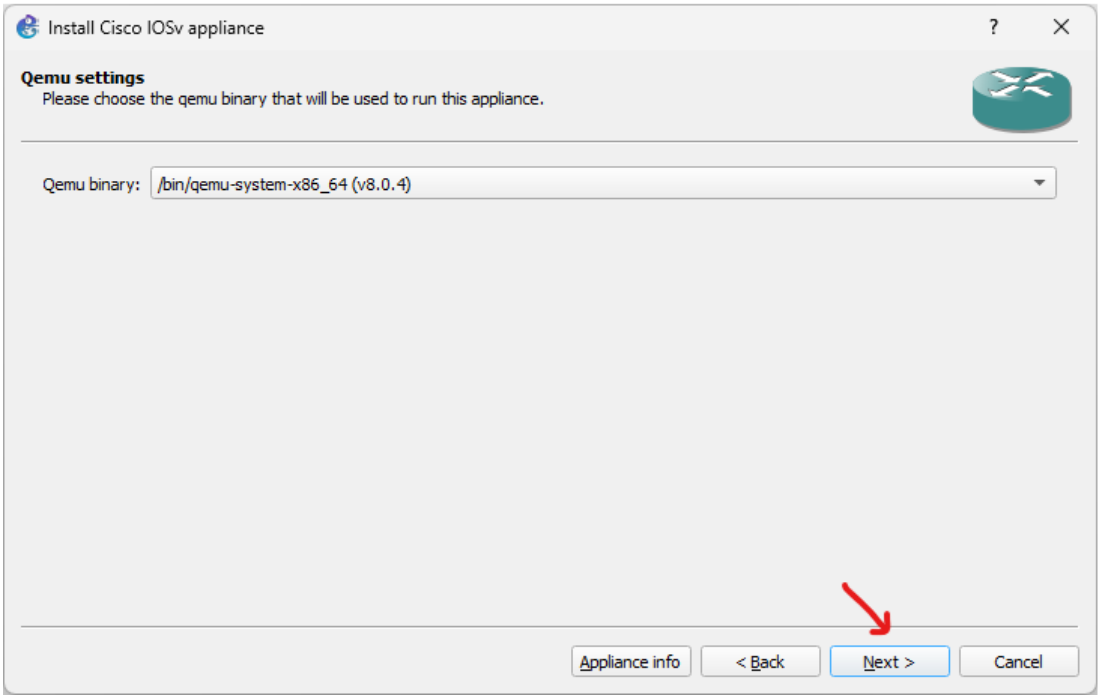
- ## Router installation

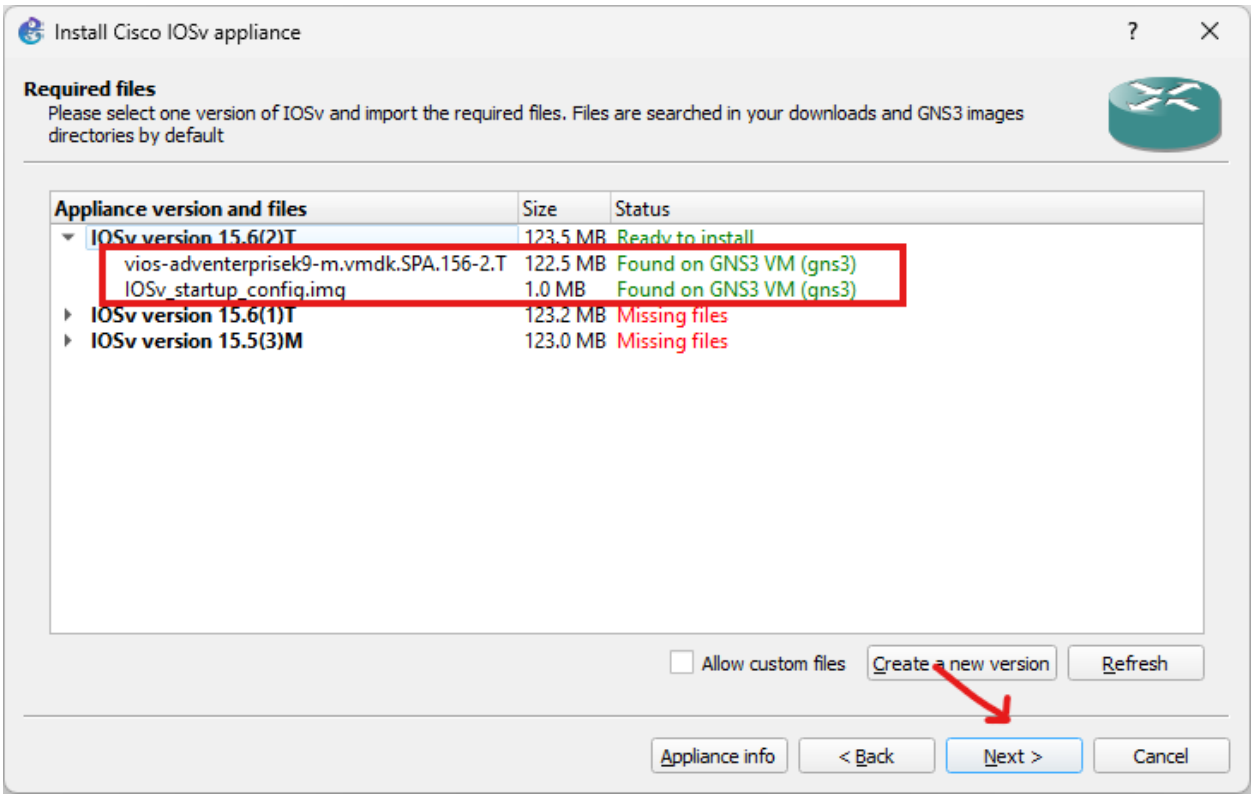
c3640-a3js-mz.124-23.image	12/13/2025 3:08 PM	IMAGE File	64,160 KB
cisco-iosv	12/12/2025 6:43 PM	GNS3 Appliance File	3 KB
cisco-iosv.gns3a.md5sum	12/12/2025 6:43 PM	MD5SUM File	1 KB
IOSv_startup_config	12/12/2025 6:42 PM	Disc Image File	1,024 KB
IOSv_startup_config.img.md5sum	12/12/2025 6:42 PM	MD5SUM File	1 KB
vios-adventerprisek9-m.vmdk.SPA.156-2	12/12/2025 6:41 PM	Perl Source File	125,440 KB
vios-adventerprisek9-m.vmdk.SPA.156-2....	12/12/2025 6:25 PM	MD5SUM File	1 KB

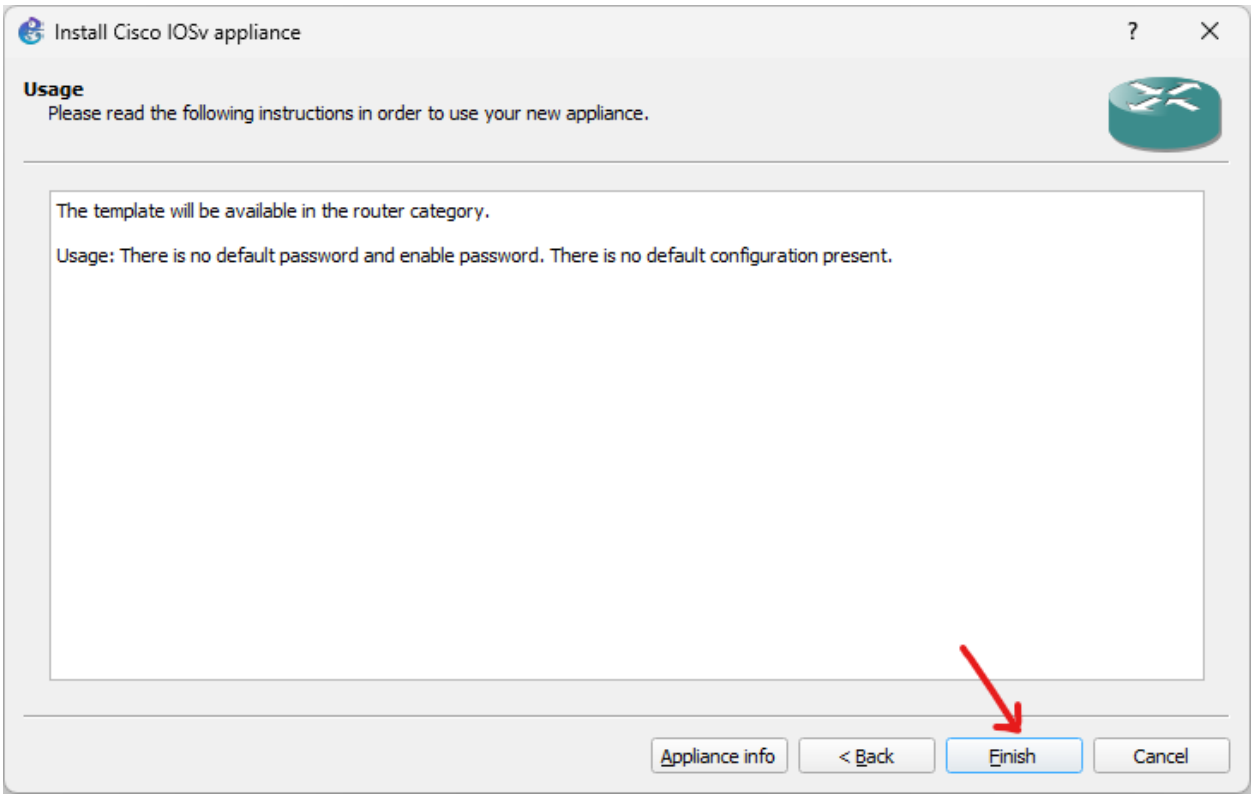


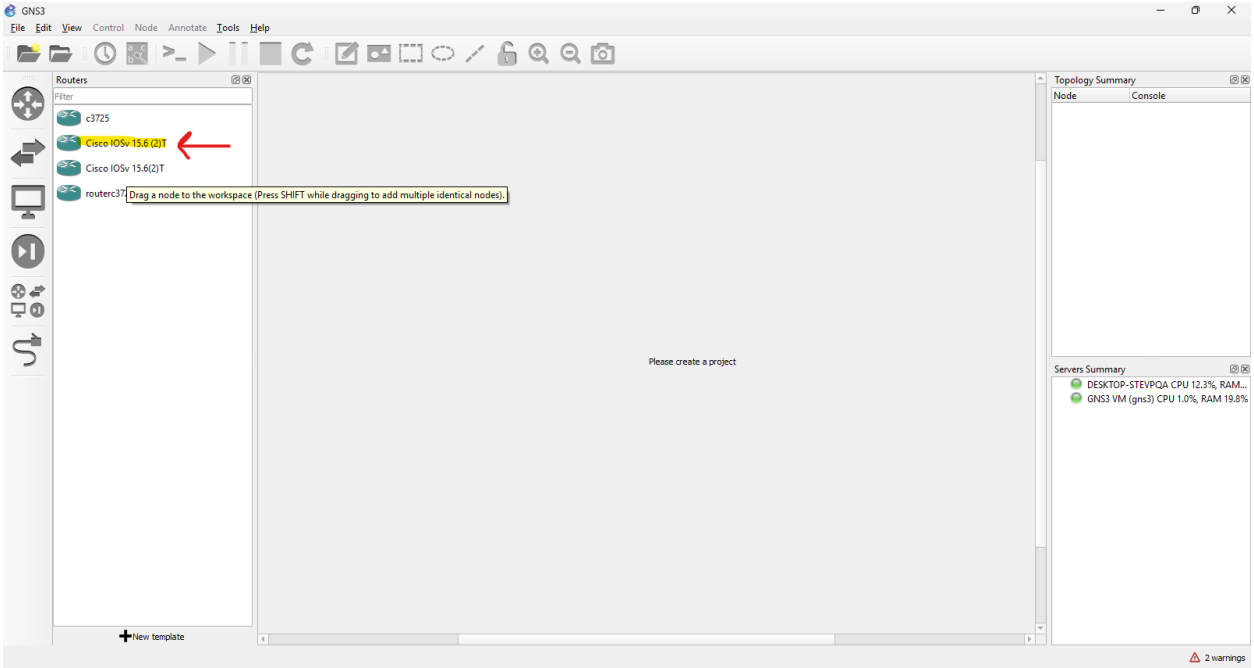

Import appliance
« BSCYS-SEM 3 › CN › images ›
Search images
Organize
New folder
Name
Date modified
Type
Size
CNproject
12/13/2025 9:32 PM
File folder
cisco-iosv
12/12/2025 6:43 PM
GNS3 Appliance File
3 KB
File name:
cisco-iosv
GNS3 Appliance (*.gns3applian
Open
Cancel


Install Cisco IOSv appliance
Server
Please choose a server type to install the appliance. The grayed out server types are not supported or configured.
Server type
Install the appliance on a remote server
Install the appliance on the GNS3 VM (recommended)
Install the appliance on your local computer
Appliance info
Next >
Cancel


Install Cisco IOSv appliance
Qemu settings
Please choose the qemu binary that will be used to run this appliance.
Qemu binary:
/bin/qemu-system-x86_64 (v8.0.4)
Appliance info
< Back
Next >
Cancel

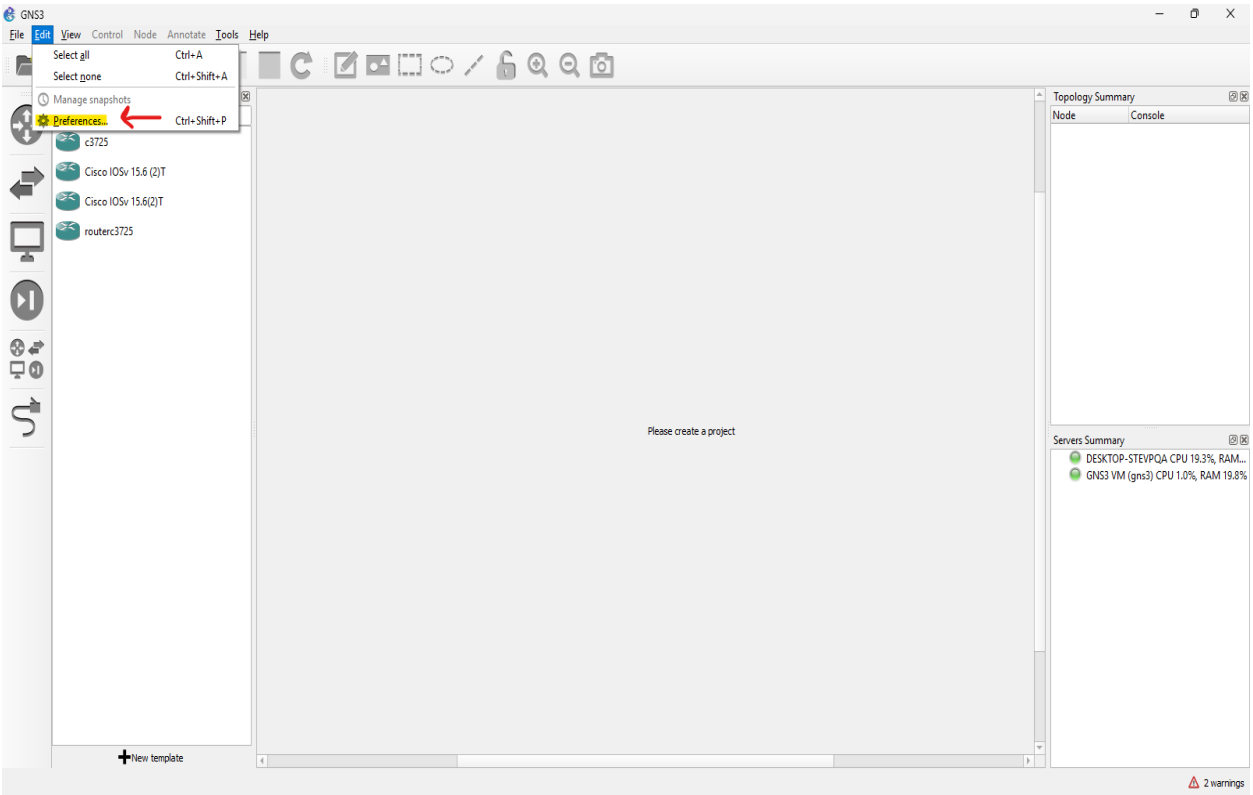

Install Cisco IOSv appliance
Required files
Please select one version of IOSv and import the required files. Files are searched in your downloads and GNS3 images directories by default
Appliance version and files
Size
Status
IOSv version 15.6(2)T
123.5 MB
Ready to install
vios-adventerprisek9-m.vmdk.SPA.156-2.T
122.5 MB
Found on GNS3 VM (gns3)
IOSv_startup_config.img
1.0 MB
Found on GNS3 VM (gns3)
IOSv version 15.6(1)T
123.2 MB
Missing files
IOSv version 15.5(3)M
123.0 MB
Missing files
Allow custom files
Create a new version
Refresh
Appliance info
< Back
Next >
Cancel

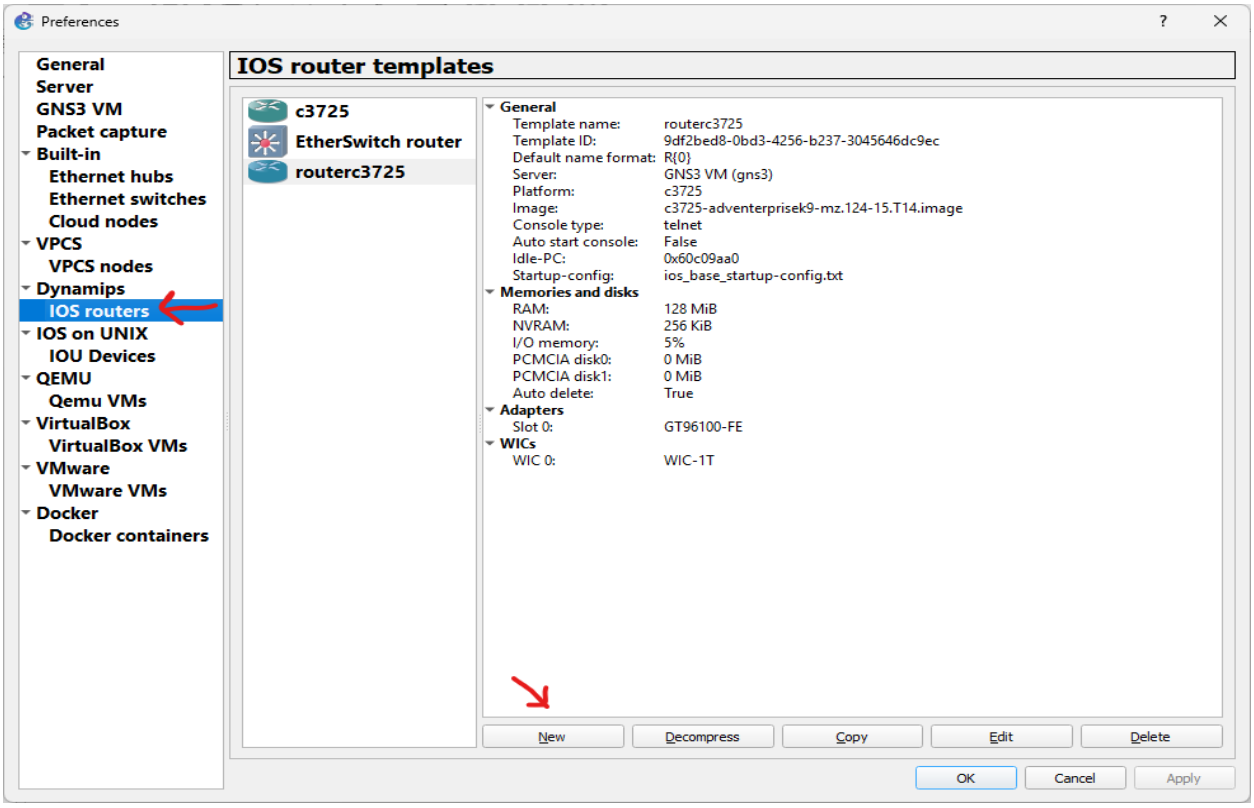

Install Cisco IOSv appliance
Usage
Please read the following instructions in order to use your new appliance.
The template will be available in the router category.
Usage: There is no default password and enable password. There is no default configuration present.
Appliance info
< Back
Finish
Cancel

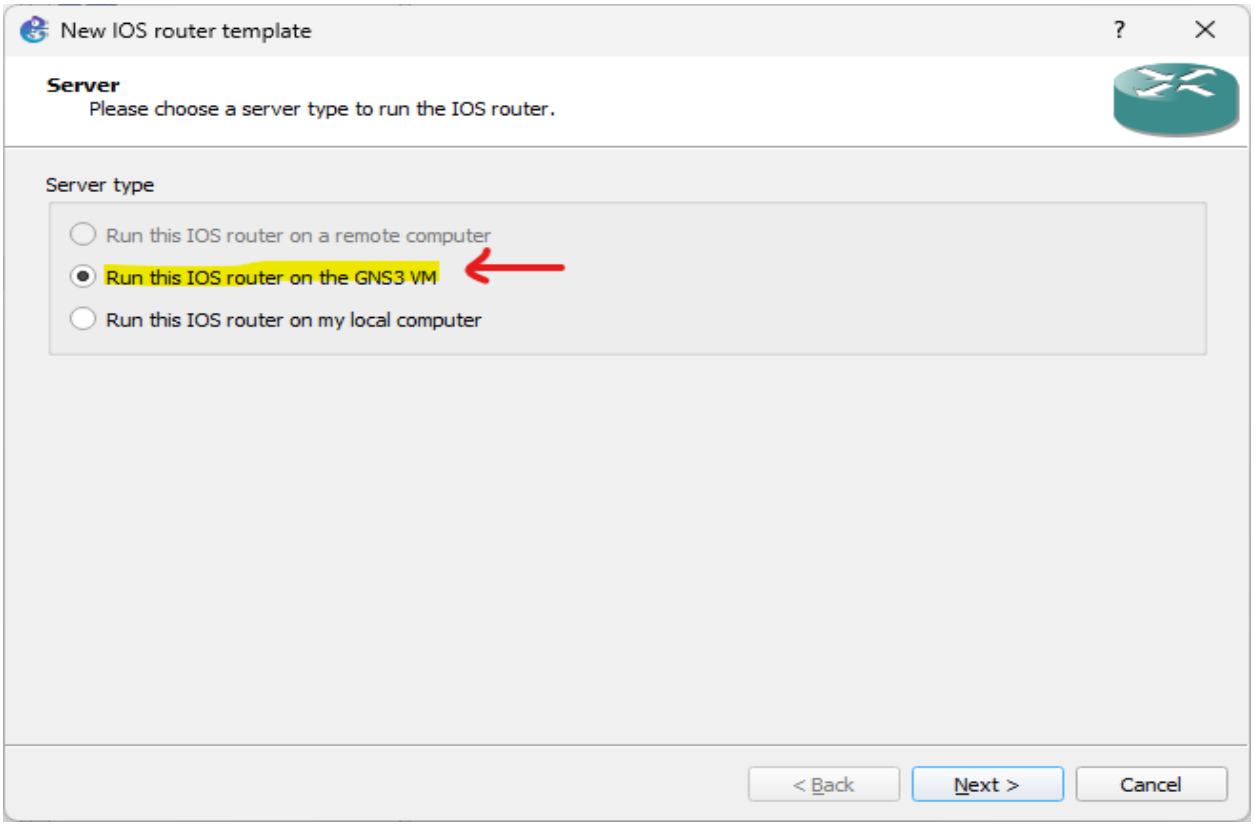

GNS3
File Edit View Control Node Annotate Tools Help
Routers
Filter
c3725
Cisco IOSv 15.6 (2)T
Cisco IOSv 15.6(2)T
routerc37
Drag a node to the workspace (Press SHIFT while dragging to add multiple identical nodes).
New template
Please create a project
Topology Summary
Node
Console
Servers Summary
DESKTOP-STEVPQA CPU 12.3%, RAM...
GNS3 VM (gns3) CPU 1.0%, RAM 19.8%
2 warnings

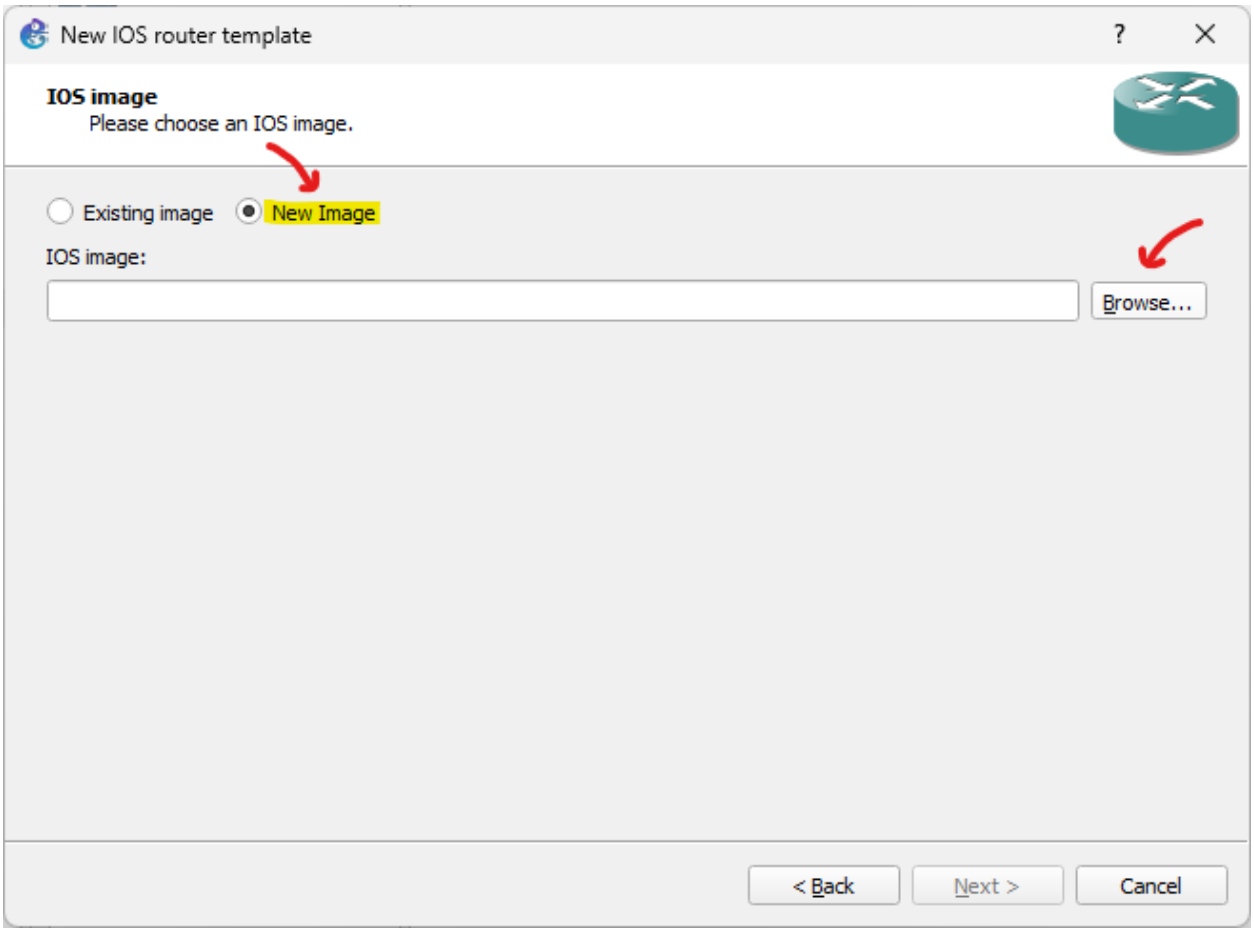
- ## Switch installation

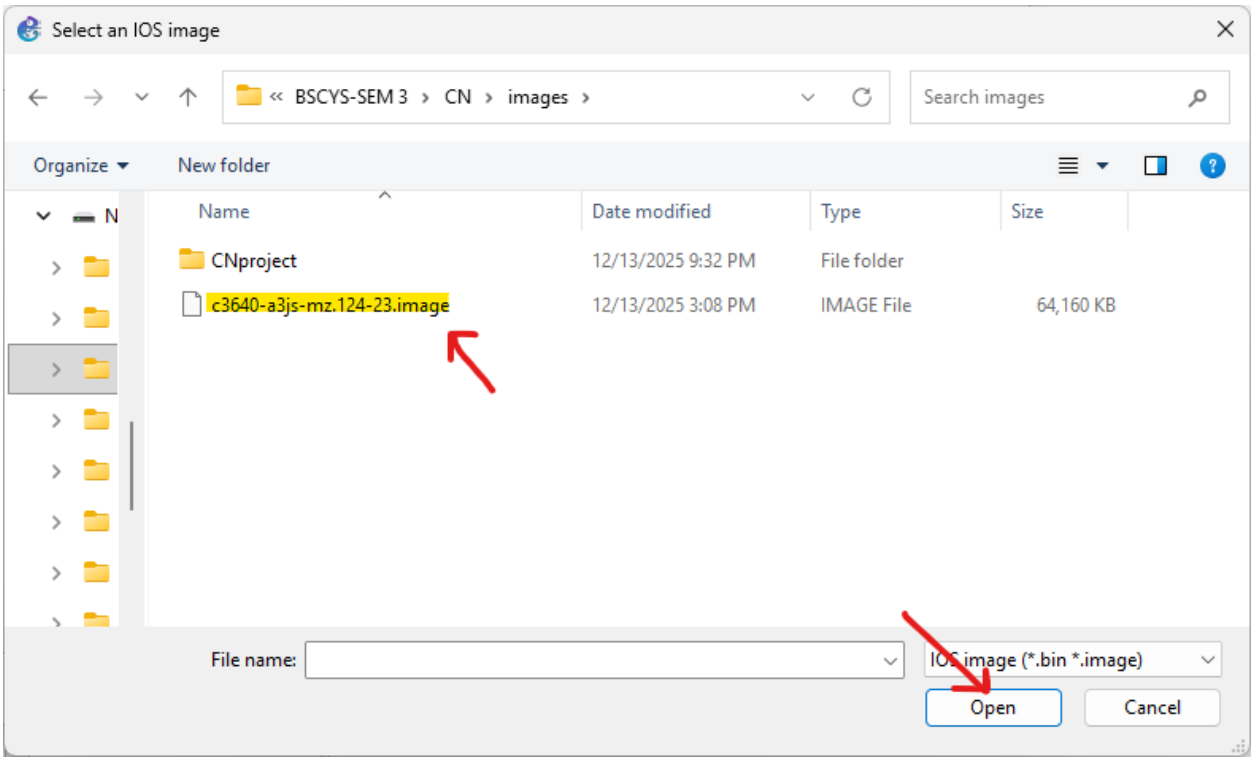
c3640-a3js-mz.124-23.image	12/13/2025 3:08 PM	IMAGE File	64,160 KB
cisco-iosv	12/12/2025 6:43 PM	GNS3 Appliance File	3 KB
cisco-iosv.gns3a.md5sum	12/12/2025 6:43 PM	MD5SUM File	1 KB
IOSv_startup_config	12/12/2025 6:42 PM	Disc Image File	1,024 KB
IOSv_startup_config.img.md5sum	12/12/2025 6:42 PM	MD5SUM File	1 KB
vios-adventerprisek9-m.vmdk.SPA.156-2	12/12/2025 6:41 PM	Perl Source File	125,440 KB
vios-adventerprisek9-m.vmdk.SPA.156-2....	12/12/2025 6:25 PM	MD5SUM File	1 KB

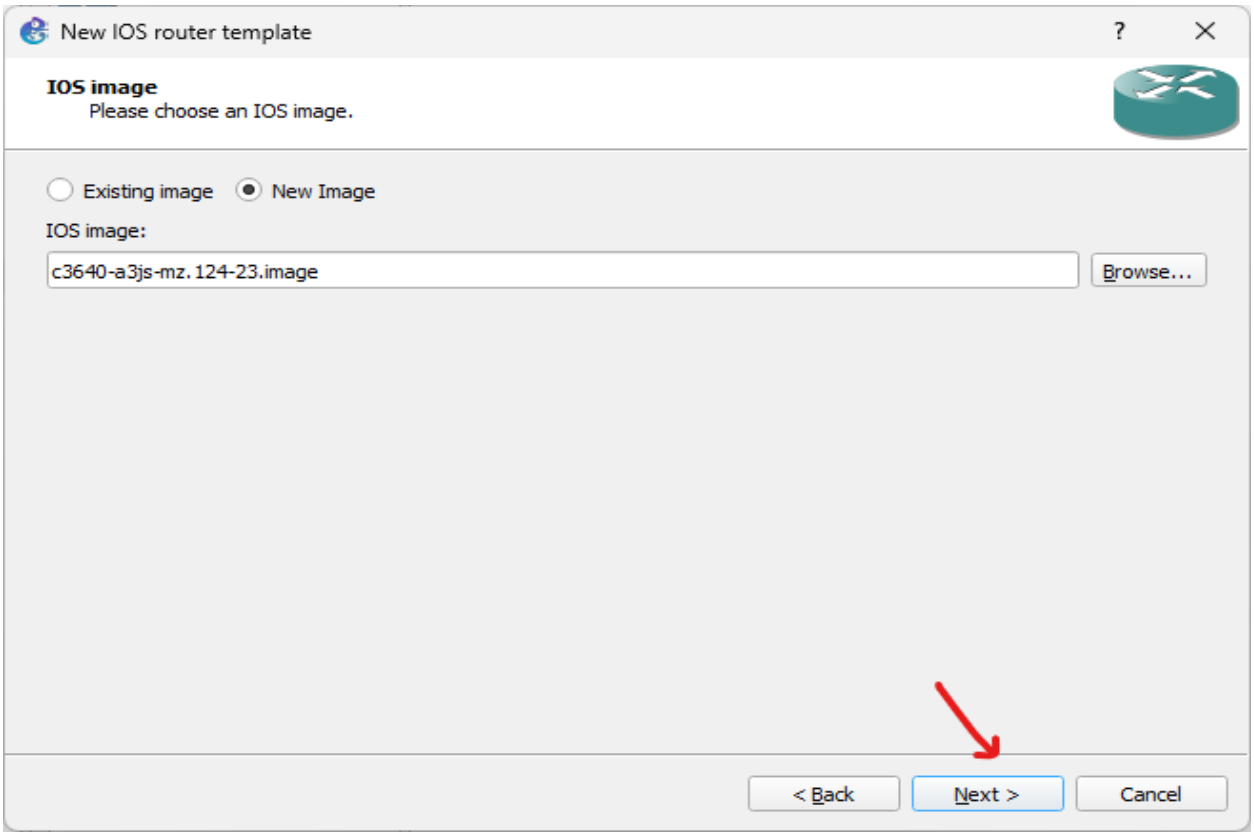


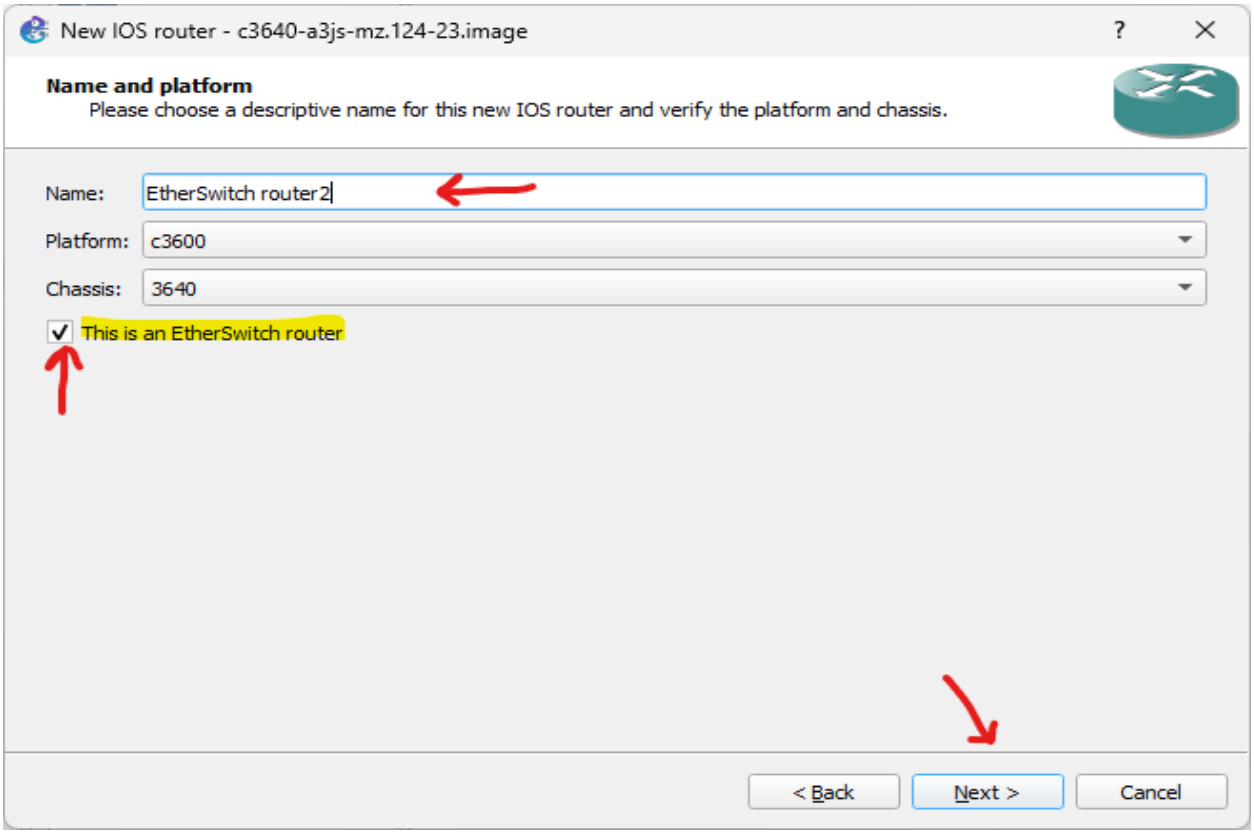

Preferences
General
Server
GNS3 VM
Packet capture
Built-in
Ethernet hubs
Ethernet switches
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containers
IOS router templates
c3725
EtherSwitch router
routerc3725
General
Template name: routerc3725
Template ID: 9df2bed8-0bd3-4256-b237-3045646dc9ec
Default name format: R{0}
Server: GNS3 VM (gns3)
Platform: c3725
Image: c3725-adventerprisek9-mz.124-15.T14.image
Console type: telnet
Auto start console: False
Idle-PC: 0x60c09aa0
Startup-config: ios_base_startup-config.txt
Memories and disks
RAM: 128 MiB
NVRAM: 256 KiB
I/O memory: 5%
PCMCIA disk0: 0 MiB
PCMCIA disk1: 0 MiB
Auto delete: True
Adapters
Slot 0: GT96100-FE
WICs
WIC 0: WIC-1T
New
Decompress
Copy
Edit
Delete
OK
Cancel
Apply

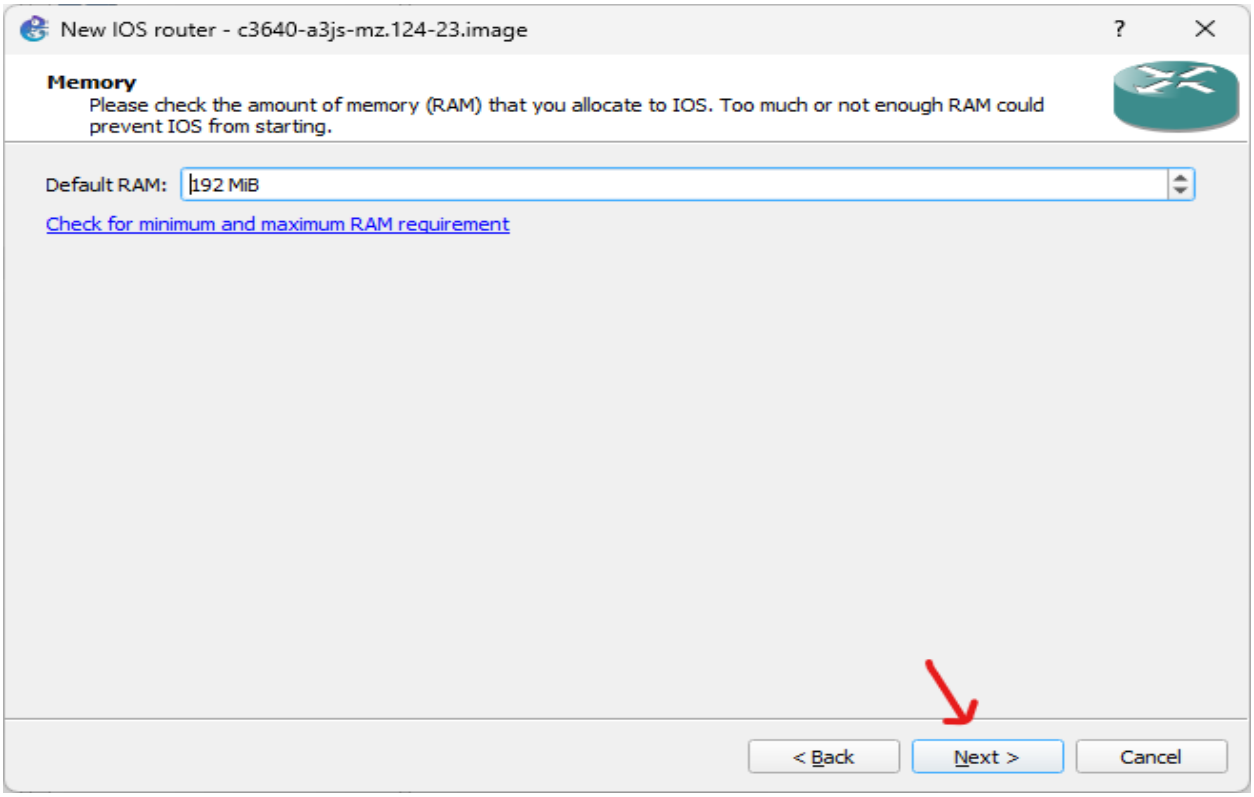

New IOS router template
Server
Please choose a server type to run the IOS router.
Server type
Run this IOS router on a remote computer
Run this IOS router on the GNS3 VM
Run this IOS router on my local computer
< Back
Next >
Cancel

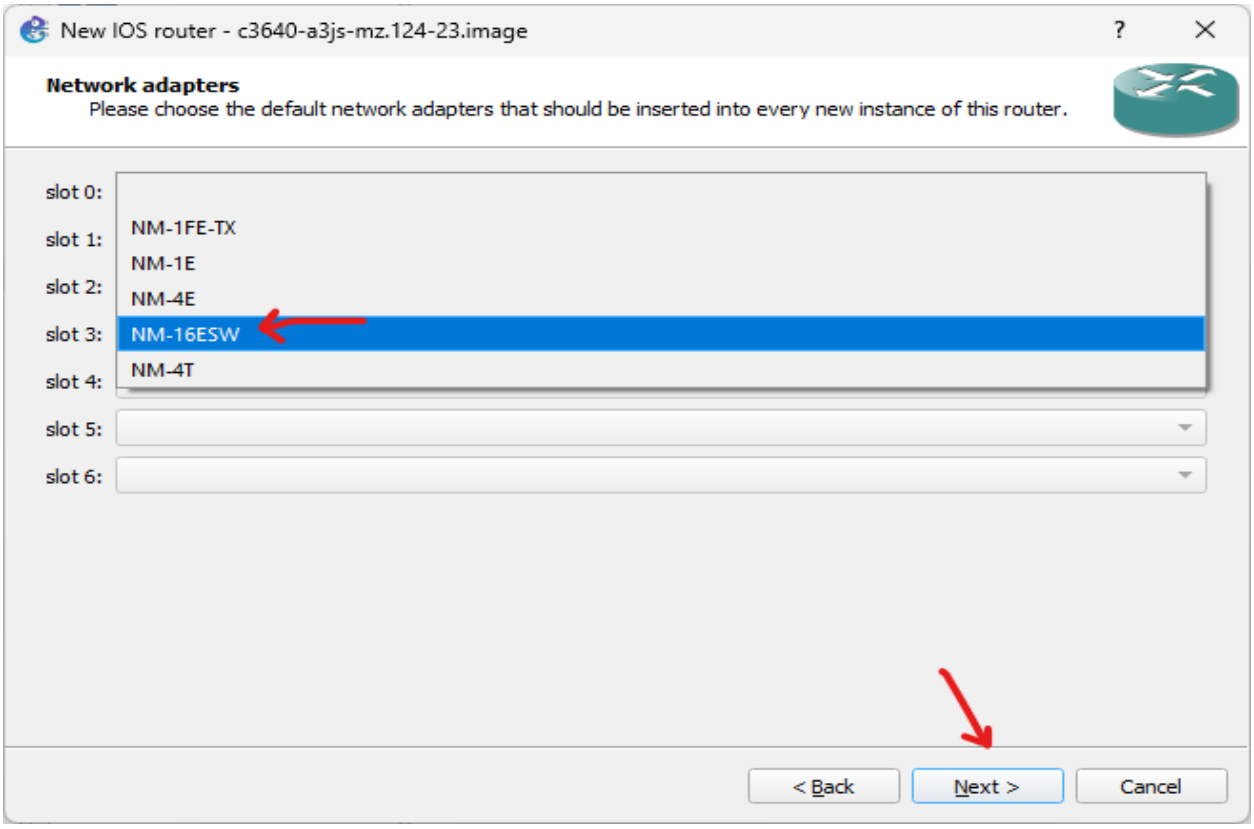

New IOS router template
IOS image
Please choose an IOS image.
Existing image
New Image
IOS image:
Browse...
< Back
Next >
Cancel

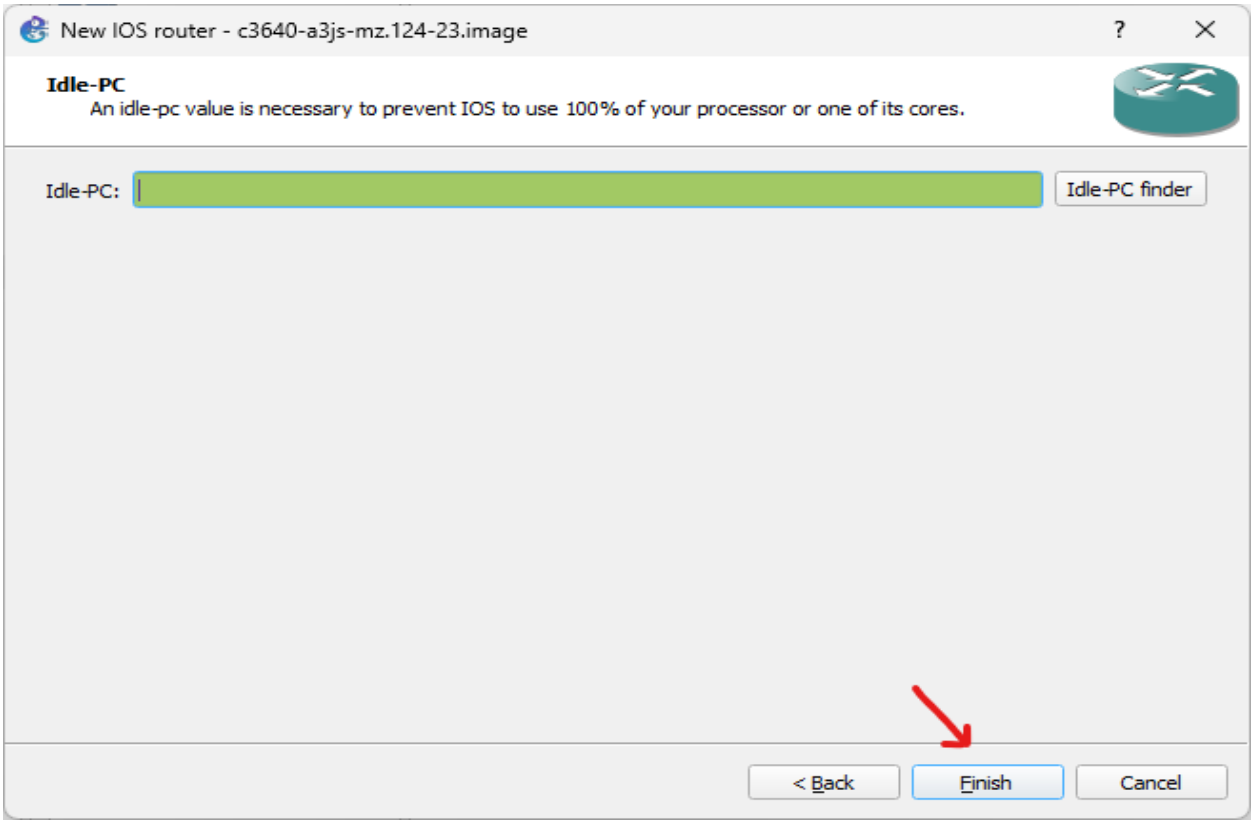

Select an IOS image
« BSCYS-SEM 3 › CN › images ›
Search images
Organize
New folder
Name
Date modified
Type
Size
CNproject
12/13/2025 9:32 PM
File folder
c3640-a3js-mz.124-23.image
12/13/2025 3:08 PM
IMAGE File
64,160 KB
File name:
IOS image (*.bin *.image)
Open
Cancel

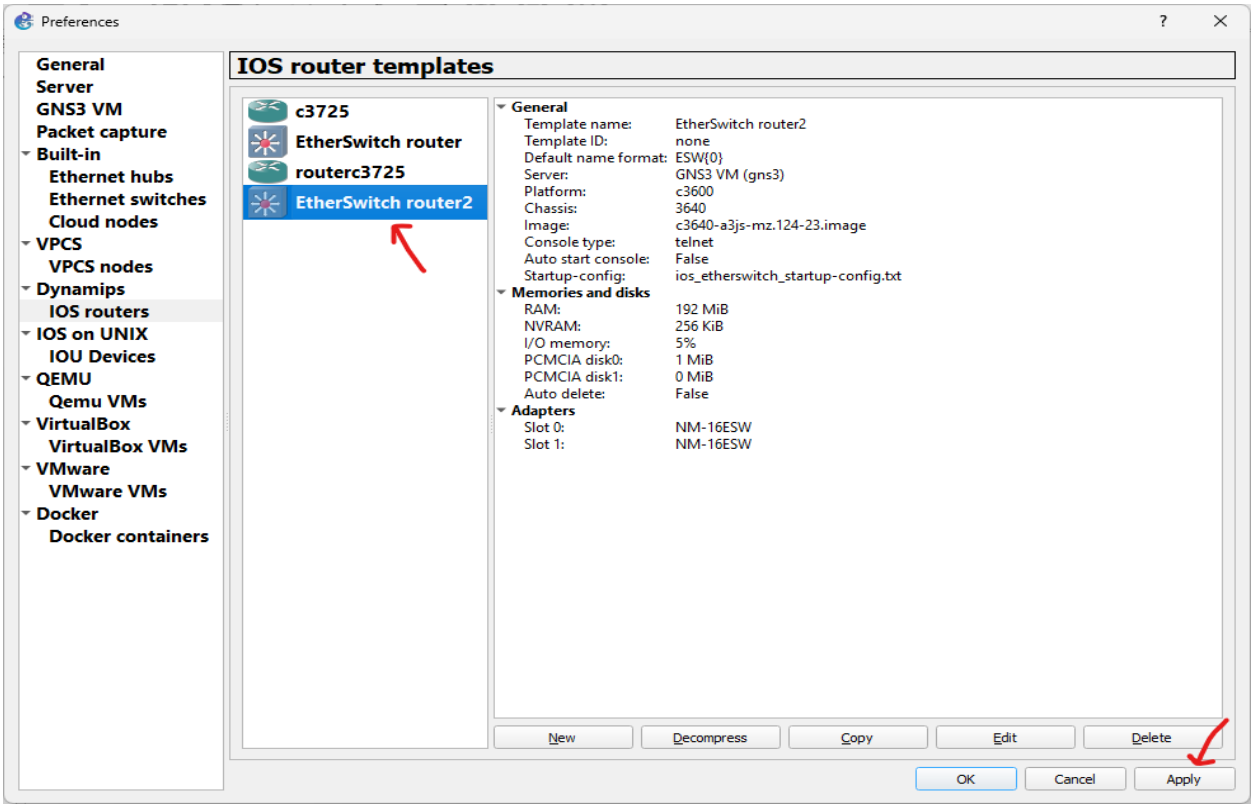

New IOS router template
IOS image
Please choose an IOS image.
Existing image
New Image
IOS image:
c3640-a3js-mz.124-23.image
Browse...
< Back
Next >
Cancel


New IOS router - c3640-a3js-mz.124-23.image
Name and platform
Please choose a descriptive name for this new IOS router and verify the platform and chassis.
Name:
EtherSwitch router2
Platform:
c3600
Chassis:
3640
This is an EtherSwitch router
< Back
Next >
Cancel


New IOS router - c3640-a3js-mz.124-23.image
Memory
Please check the amount of memory (RAM) that you allocate to IOS. Too much or not enough RAM could prevent IOS from starting.
Default RAM:
192 MiB
Check for minimum and maximum RAM requirement
< Back
Next >
Cancel


New IOS router - c3640-a3js-mz.124-23.image
Network adapters
Please choose the default network adapters that should be inserted into every new instance of this router.
slot 0:
slot 1:
slot 2:
slot 3:
slot 4:
slot 5:
slot 6:
NM-1FE-TX
NM-1E
NM-4E
NM-16ESW
NM-4T
< Back
Next >
Cancel


New IOS router - c3640-a3js-mz.124-23.image
Idle-PC
An idle-pc value is necessary to prevent IOS to use 100% of your processor or one of its cores.
Idle-PC:
Idle-PC finder
< Back
Finish
Cancel


Preferences
General
Server
GNS3 VM
Packet capture
Built-in
Ethernet hubs
Ethernet switches
Cloud nodes
VPCS
VPCS nodes
Dynamips
IOS routers
IOS on UNIX
IOU Devices
QEMU
Qemu VMs
VirtualBox
VirtualBox VMs
VMware
VMware VMs
Docker
Docker containers
IOS router templates
c3725
EtherSwitch router
routerc3725
EtherSwitch router2
General
Template name: EtherSwitch router2
Template ID: none
Default name format: ESW{0}
Server: GNS3 VM (gns3)
Platform: c3600
Chassis: 3640
Image: c3640-a3js-mz.124-23.image
Console type: telnet
Auto start console: False
Startup-config: ios_etherswitch_startup-config.txt
Memories and disks
RAM: 192 MiB
NVRAM: 256 KiB
I/O memory: 5%
PCMCIA disk0: 1 MiB
PCMCIA disk1: 0 MiB
Auto delete: False
Adapters
Slot 0: NM-16ESW
Slot 1: NM-16ESW
New
Decompress
Copy
Edit
Delete
OK
Cancel
Apply